

Linear Algebra 2

Problem 1 — Eigenvalues of an upper triangular matrix

Exact statement. A square matrix A is *upper triangular* if $A_{ij} = 0$ for $i > j$. Show that the eigenvalues of such a matrix are its diagonal entries:

$$\lambda_i = A_{ii} \quad (\text{no sum over } i).$$

Solution. The characteristic polynomial is

$$p_A(t) = \det(A - tI).$$

If A is upper triangular, then $A - tI$ is also upper triangular with diagonal entries $A_{11} - t, \dots, A_{nn} - t$. The determinant of a triangular matrix equals the product of its diagonal entries, hence

$$p_A(t) = \prod_{i=1}^n (A_{ii} - t).$$

Therefore the roots are $t = A_{ii}$, so the eigenvalues are exactly the diagonal entries.

Problem 2 — One eigenvalue, not diagonalizable

Exact statement. Show that

$$M = \begin{pmatrix} 0 & 1 & 0 \\ -4 & 4 & 0 \\ -2 & 1 & 2 \end{pmatrix}$$

has characteristic equation $(t - 2)^3 = 0$. Explain, as simply as possible, why M is not diagonalizable.

Solution. Compute

$$tI - M = \begin{pmatrix} t & -1 & 0 \\ 4 & t - 4 & 0 \\ 2 & -1 & t - 2 \end{pmatrix}.$$

Expanding along the third column gives

$$\det(tI - M) = (t - 2) \det \begin{pmatrix} t & -1 \\ 4 & t - 4 \end{pmatrix} = (t - 2)(t(t - 4) + 4) = (t - 2)(t^2 - 4t + 4) = (t - 2)^3.$$

Thus 2 is the only eigenvalue, with algebraic multiplicity 3.

To test diagonalizability, compute the eigenspace:

$$M - 2I = \begin{pmatrix} -2 & 1 & 0 \\ -4 & 2 & 0 \\ -2 & 1 & 0 \end{pmatrix}.$$

All rows are multiples of $(-2, 1, 0)$, so $\text{rank}(M - 2I) = 1$, hence

$$\dim \ker(M - 2I) = 3 - 1 = 2.$$

The geometric multiplicity is $2 < 3$, so M does not have 3 linearly independent eigenvectors and is not diagonalizable.

Problem 3 — Completing an orthogonal matrix

Exact statement. Find a, b, c such that

$$Q = \begin{pmatrix} \frac{1}{3} & 0 & a \\ \frac{2}{3} & \frac{1}{\sqrt{2}} & b \\ \frac{2}{3} & -\frac{1}{\sqrt{2}} & c \end{pmatrix}$$

is an orthogonal matrix. Does this condition determine a, b, c uniquely?

Solution. Let the columns be

$$v_1 = \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}, \quad v_3 = \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

One checks

$$\|v_1\|^2 = \frac{1}{9} + \frac{4}{9} + \frac{4}{9} = 1, \quad \|v_2\|^2 = \frac{1}{2} + \frac{1}{2} = 1, \quad v_1 \cdot v_2 = 0,$$

so v_1, v_2 are already orthonormal. Thus v_3 must be a unit vector orthogonal to both v_1 and v_2 , hence

$$v_3 = \pm(v_1 \times v_2).$$

Compute

$$v_1 \times v_2 = \frac{\sqrt{2}}{6} \begin{pmatrix} -4 \\ 1 \\ 1 \end{pmatrix}.$$

Therefore

$$(a, b, c) = \left(-\frac{2\sqrt{2}}{3}, \frac{\sqrt{2}}{6}, \frac{\sqrt{2}}{6} \right) \quad \text{or} \quad (a, b, c) = \left(\frac{2\sqrt{2}}{3}, -\frac{\sqrt{2}}{6}, -\frac{\sqrt{2}}{6} \right).$$

So the condition does not determine a, b, c uniquely: there are exactly two choices (sign $\pm$).

Problem 4 — Spectral decomposition and A^{-1}

Exact statement. Determine the eigenvalues and eigenvectors of the symmetric matrix

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}.$$

Use an identity of the form $P^T A P = D$, where D is diagonal, to find A^{-1} .

Solution. A set of eigenpairs is:

$$\lambda_1 = 1, v_1 = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}, \quad \lambda_2 = 2, v_2 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \quad \lambda_3 = 4, v_3 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

These eigenvectors are mutually orthogonal. Normalize:

$$u_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}, \quad u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \quad u_3 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

Let $P = [u_1 \ u_2 \ u_3]$ (orthogonal) and $D = \text{diag}(1, 2, 4)$. Then

$$P^T A P = D.$$

Invert:

$$A^{-1} = P D^{-1} P^T, \quad D^{-1} = \text{diag}\left(1, \frac{1}{2}, \frac{1}{4}\right).$$

A direct computation yields

$$A^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{5}{8} & \frac{1}{8} \\ -\frac{1}{4} & \frac{1}{8} & \frac{5}{8} \end{pmatrix}.$$

Problem 5 — Diagonalising a rotated quadratic form

Exact statement. Diagonalise the quadratic form in $\mathbb{R}^2$ defined by

$$F(x, y) = (a \cos^2 \theta + b \sin^2 \theta)x^2 + 2(a - b)(\sin \theta \cos \theta)xy + (a \sin^2 \theta + b \cos^2 \theta)y^2,$$

i.e. find its eigenvalues and principal axes (a, b, θ are constants).

Solution. Write $F(x, y) = \begin{pmatrix} x & y \end{pmatrix} Q \begin{pmatrix} x \\ y \end{pmatrix}$ with

$$Q = \begin{pmatrix} a \cos^2 \theta + b \sin^2 \theta & (a - b) \sin \theta \cos \theta \\ (a - b) \sin \theta \cos \theta & a \sin^2 \theta + b \cos^2 \theta \end{pmatrix}.$$

Let

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Then one checks

$$Q = R \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} R^T.$$

Hence the eigenvalues are a and b . The principal axes are the rotated directions

$$(\cos \theta, \sin \theta) \quad \text{and} \quad (-\sin \theta, \cos \theta).$$

Equivalently, in rotated coordinates

$$\begin{pmatrix} u \\ v \end{pmatrix} = R^T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \cos \theta + y \sin \theta \\ -x \sin \theta + y \cos \theta \end{pmatrix},$$

the quadratic form becomes

$$F(x, y) = a u^2 + b v^2.$$

Problem 6 — Anti-hermitian and unitary matrices: eigenvalues and orthogonality

Exact statement. (i) A matrix A is anti-hermitian, $A^\dagger = -A$; show that the eigenvalues of A are pure-imaginary. (ii) A matrix U is unitary, $U^\dagger U = I$; show that the eigenvalues of U have unit modulus. (iii) In each of the cases (i) and (ii), show that eigenvectors with distinct eigenvalues are orthogonal.

Solution. (i) Let $Av = \lambda v$ with $v \neq 0$. Then

$$v^\dagger Av = \lambda v^\dagger v.$$

Taking complex conjugates,

$$(v^\dagger Av)^* = v^\dagger A^\dagger v = v^\dagger (-A)v = -v^\dagger Av.$$

Thus $z := v^\dagger Av$ satisfies $z^* = -z$, hence z is purely imaginary. Since $v^\dagger v > 0$ is real,

$$\lambda = \frac{v^\dagger Av}{v^\dagger v}$$

is purely imaginary.

(ii) If $Uv = \lambda v$, then

$$v^\dagger v = v^\dagger U^\dagger U v = (Uv)^\dagger (Uv) = (\lambda v)^\dagger (\lambda v) = |\lambda|^2 v^\dagger v.$$

Since $v \neq 0$, we get $|\lambda| = 1$.

(iii) For the anti-hermitian case, let $Av = \lambda v$ and $Aw = \mu w$. Then

$$v^\dagger Aw = \mu v^\dagger w,$$

but also

$$v^\dagger Aw = (A^\dagger v)^\dagger w = (-Av)^\dagger w = -(\lambda v)^\dagger w = -\bar{\lambda} v^\dagger w.$$

Hence $(\mu + \bar{\lambda})v^\dagger w = 0$. By (i), $\lambda = i\alpha$ and $\mu = i\beta$ with $\alpha, \beta \in \mathbb{R}$, so $\mu + \bar{\lambda} = i(\beta - \alpha)$. If $\lambda \neq \mu$ then $\beta \neq \alpha$, hence $v^\dagger w = 0$.

For the unitary case, if $Uv = \lambda v$ then applying $U^\dagger$ gives $v = \lambda U^\dagger v$, hence $U^\dagger v = \lambda^{-1}v = \bar{\lambda}v$ (since $|\lambda| = 1$). Now with $Uw = \mu w$,

$$v^\dagger Uw = \mu v^\dagger w, \quad v^\dagger Uw = (U^\dagger v)^\dagger w = (\bar{\lambda}v)^\dagger w = \lambda v^\dagger w.$$

Thus $(\mu - \lambda)v^\dagger w = 0$, and if $\mu \neq \lambda$ then $v^\dagger w = 0$.

Problem 7 — Cayley–Hamilton for 2×2 , powers of a specific matrix

Exact statement. Check, by direct calculation, that the Cayley–Hamilton Theorem holds for a general 2×2 matrix. Find the characteristic polynomial for

$$A = \begin{pmatrix} 3 & 4 \\ -1 & -1 \end{pmatrix}$$

and deduce that $A^2 = 2A - I$. Is A diagonalisable? Show by induction that

$$A^n = \alpha_n A + \beta_n I, \quad n \geq 0,$$

for real numbers α_n and β_n . Solve the recurrences satisfied by α_n and β_n and hence find A^n explicitly.

Solution. (a) General 2×2 case. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \operatorname{tr}(A) = a + d, \quad \det(A) = ad - bc.$$

Then

$$p_A(t) = \det(tI - A) = t^2 - (a + d)t + (ad - bc).$$

Cayley–Hamilton claims

$$p_A(A) = A^2 - (\operatorname{tr} A)A + (\det A)I = 0.$$

Compute

$$A^2 = \begin{pmatrix} a^2 + bc & ab + bd \\ ac + dc & bc + d^2 \end{pmatrix}.$$

Then entrywise one checks:

$$a^2 + bc - (a + d)a + (ad - bc) = 0, \quad ab + bd - (a + d)b = 0,$$

$$ac + dc - (a + d)c = 0, \quad bc + d^2 - (a + d)d + (ad - bc) = 0,$$

so indeed $p_A(A) = 0$.

(b) The given matrix. For

$$A = \begin{pmatrix} 3 & 4 \\ -1 & -1 \end{pmatrix}$$

we have $\operatorname{tr}(A) = 2$ and $\det(A) = 1$, hence

$$p_A(t) = t^2 - 2t + 1 = (t - 1)^2.$$

By Cayley–Hamilton,

$$A^2 - 2A + I = 0 \implies A^2 = 2A - I.$$

The only eigenvalue is 1 (algebraic multiplicity 2). Since

$$A - I = \begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix}$$

has rank 1, the eigenspace has dimension 1, so A is not diagonalizable.

(c) **Powers.** Set $A^n = \alpha_n A + \beta_n I$. From $A^0 = I$ and $A^1 = A$ we have

$$(\alpha_0, \beta_0) = (0, 1), \quad (\alpha_1, \beta_1) = (1, 0).$$

Assuming $A^n = \alpha_n A + \beta_n I$, multiply by A :

$$A^{n+1} = \alpha_n A^2 + \beta_n A = \alpha_n (2A - I) + \beta_n A = (2\alpha_n + \beta_n)A - \alpha_n I.$$

Thus

$$\alpha_{n+1} = 2\alpha_n + \beta_n, \quad \beta_{n+1} = -\alpha_n.$$

Eliminating β_n via $\beta_n = -\alpha_{n-1}$ gives

$$\alpha_{n+1} = 2\alpha_n - \alpha_{n-1}, \quad \alpha_0 = 0, \alpha_1 = 1.$$

This yields $\alpha_n = n$, and then $\beta_n = -\alpha_{n-1} = 1 - n$. Therefore

$$\boxed{A^n = nA + (1 - n)I, \quad n \geq 0.}$$

Problem 8 — Matrices of a linear map in nonstandard bases (and change of basis)

Exact statement. Define the $m \times n$ matrix A that represents a linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with respect to general bases $\{e_1, \dots, e_n\}$ and $\{f_1, \dots, f_m\}$.

(a) Taking $n = 2, m = 3$, let T be defined by

$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}, \quad T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 0 \\ 3 \end{pmatrix}.$$

Find the matrix A with respect to the bases

$$e_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad e_2 = \begin{pmatrix} -3 \\ 2 \end{pmatrix}; \quad f_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad f_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad f_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

(b) Taking $n = m = 3$, let T be reflection in the plane $x_1 \sin \theta = x_2 \cos \theta$. Find the matrix A with respect to a convenient choice of bases (to be specified) such that $e_i = f_i$ ($i = 1, 2, 3$).

(c) Taking $n = m = 2$, let T be the shear with parameter λ defined by

$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \lambda \\ 1 \end{pmatrix}.$$

Find the matrix A when $e_1 = f_1$ and $e_2 = f_2$ are the standard basis vectors for $\mathbb{R}^2$; find also the matrix A' with respect to the new basis $e'_1 = f'_1 = -e_2$ and $e'_2 = f'_2 = e_1$. Show that $A' = R^{-1}AR$ for a rotation matrix R and comment on this result.

Solution. General definition. The representing matrix $A = (a_{ij})$ is defined by

$$T(e_j) = \sum_{i=1}^m a_{ij} f_i \quad (j = 1, \dots, n).$$

Equivalently, for all $v \in \mathbb{R}^n$,

$$[T(v)]_F = A[v]_E.$$

(a) In the standard basis of $\mathbb{R}^2$, the matrix of T is

$$M = \begin{pmatrix} 2 & 7 \\ 1 & 0 \\ 5 & 3 \end{pmatrix}.$$

Since (f_1, f_2, f_3) is the standard basis of $\mathbb{R}^3$, coordinates in F are just components. Compute

$$\begin{aligned} T(e_1) &= T\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} - \begin{pmatrix} 7 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} -5 \\ 1 \\ 2 \end{pmatrix}, \\ T(e_2) &= T\begin{pmatrix} -3 \\ 2 \end{pmatrix} = -3 \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} + 2 \begin{pmatrix} 7 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 8 \\ -3 \\ -9 \end{pmatrix}. \end{aligned}$$

Hence the representing matrix (columns are $[T(e_1)]_F$ and $[T(e_2)]_F$) is

$$A = \begin{pmatrix} -5 & 8 \\ 1 & -3 \\ 2 & -9 \end{pmatrix}.$$

(b) The reflecting plane is $x_1 \sin \theta - x_2 \cos \theta = 0$, so a unit normal is

$$e_1 = n = (\sin \theta, -\cos \theta, 0).$$

Choose a basis adapted to the plane:

$$e_2 = (\cos \theta, \sin \theta, 0) \in \text{plane}, \quad e_3 = (0, 0, 1) \in \text{plane}.$$

Reflection fixes vectors in the plane and flips the normal, hence

$$T(e_1) = -e_1, \quad T(e_2) = e_2, \quad T(e_3) = e_3.$$

With respect to the basis (e_1, e_2, e_3) (and taking $f_i = e_i$), the matrix is

$$A = \text{diag}(-1, 1, 1).$$

(Optionally, in the standard basis one may write $T(x) = x - 2(x \cdot n)n$ to obtain $\begin{pmatrix} \cos 2\theta & \sin 2\theta & 0 \\ \sin 2\theta & -\cos 2\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$.)

(c) In the standard basis, the columns are $T(e_1)$ and $T(e_2)$, so

$$A = \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}.$$

For the new basis $e'_1 = -e_2$ and $e'_2 = e_1$, the change-of-basis matrix is

$$R = [e'_1 \ e'_2] = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad R^{-1} = R^T.$$

Therefore the representing matrix in the primed basis is

$$A' = R^{-1}AR,$$

and a direct multiplication gives

$$A' = \begin{pmatrix} 1 & 0 \\ -\lambda & 1 \end{pmatrix}.$$

This illustrates the general rule: matrices of the same linear map in different bases are similar, so they have the same invariants (eigenvalues, trace, determinant).

Problem 9 — Change of basis to a Jordan block; compute A^n

Exact statement. The linear map $S : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is defined in terms of its matrix A with respect to the standard basis by

$$S \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -5x + 9y \\ -4x + 7y \end{pmatrix}.$$

Find the matrix B for S with respect to the basis

$$\left\{ \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}.$$

Show that $B^n - I = n(B - I)$ for all positive integers n , and hence determine A^n . Verify that $\det(A^n) = (\det A)^n$.

Solution. From the definition,

$$A = \begin{pmatrix} -5 & 9 \\ -4 & 7 \end{pmatrix}.$$

Let $v_1 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and set

$$P = [v_1 \ v_2] = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}.$$

Compute

$$Av_1 = \begin{pmatrix} -15 + 18 \\ -12 + 14 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} = v_1, \quad Av_2 = \begin{pmatrix} -5 + 9 \\ -4 + 7 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}.$$

Write $Av_2 = av_1 + bv_2$:

$$\begin{pmatrix} 4 \\ 3 \end{pmatrix} = a \begin{pmatrix} 3 \\ 2 \end{pmatrix} + b \begin{pmatrix} 1 \\ 1 \end{pmatrix} \implies \begin{cases} 3a + b = 4 \\ 2a + b = 3 \end{cases} \implies a = 1, \ b = 1.$$

Hence the columns of B are $(1, 0)^T$ and $(1, 1)^T$, so

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Now $B = I + N$ with $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $N^2 = 0$, so

$$B^n = (I + N)^n = I + nN = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}.$$

Therefore

$$\boxed{B^n - I = n(B - I).}$$

Since $A = PBP^{-1}$, we have $A^n = PB^nP^{-1}$. Equivalently (by similarity),

$$A^n = I + n(A - I) = nA + (1 - n)I.$$

Thus

$$\boxed{A^n = n \begin{pmatrix} -5 & 9 \\ -4 & 7 \end{pmatrix} + (1 - n) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 - 6n & 9n \\ -4n & 1 + 6n \end{pmatrix}.}$$

Finally,

$$\det A = (-5)(7) - 9(-4) = 1 \quad \Rightarrow \quad \det(A^n) = 1 = (\det A)^n.$$

Problem 10 — Orthonormal eigenbases and quadric surfaces

Exact statement. Find all eigenvalues, and an orthonormal set of eigenvectors, of

$$A = \begin{pmatrix} 5 & 0 & \sqrt{3} \\ 0 & 3 & 0 \\ \sqrt{3} & 0 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}.$$

Hence sketch the surfaces

$$5x^2 + 3y^2 + 3z^2 + 2\sqrt{3}xz = 1, \quad x^2 + y^2 + z^2 - xy - yz - zx = 1.$$

Solution. (A) The eigenvalues are

$$\lambda = 2, 3, 6.$$

An orthonormal eigenbasis is

$$u_2 = \begin{pmatrix} -\frac{1}{2} \\ 0 \\ \frac{\sqrt{3}}{2} \end{pmatrix}, \quad u_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_6 = \begin{pmatrix} \frac{\sqrt{3}}{2} \\ 0 \\ \frac{1}{2} \end{pmatrix},$$

corresponding to $\lambda = 2, 3, 6$ respectively. Let

$$U = u_2 \cdot (x, y, z) = \frac{-x + \sqrt{3}z}{2}, \quad V = u_3 \cdot (x, y, z) = y, \quad W = u_6 \cdot (x, y, z) = \frac{\sqrt{3}x + z}{2}.$$

Then

$$(x, y, z)A(x, y, z)^T = 2U^2 + 3V^2 + 6W^2,$$

so the surface $5x^2 + 3y^2 + 3z^2 + 2\sqrt{3}xz = 1$ is the ellipsoid

$$2U^2 + 3V^2 + 6W^2 = 1,$$

with semi-axis lengths $1/\sqrt{2}$, $1/\sqrt{3}$, $1/\sqrt{6}$ (a rotation about the y -axis).

(B) The eigenvalues of B are

$$0 \text{ (mult. 1),} \quad 3 \text{ (mult. 2).}$$

An orthonormal eigenbasis is

$$v_0 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad v_{3,1} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad v_{3,2} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}.$$

If (α, β, γ) are coordinates in this basis then

$$(x, y, z)B(x, y, z)^T = 3\beta^2 + 3\gamma^2.$$

Moreover,

$$x^2 + y^2 + z^2 - xy - yz - zx = \frac{1}{2}(x, y, z)B(x, y, z)^T,$$

so the surface $x^2 + y^2 + z^2 - xy - yz - zx = 1$ becomes

$$\frac{1}{2}(3\beta^2 + 3\gamma^2) = 1 \iff \beta^2 + \gamma^2 = \frac{2}{3},$$

with α free. Hence it is a circular cylinder of radius $\sqrt{2/3}$ whose axis is the line spanned by $(1, 1, 1)$.

Problem 11 — Diagonalise a quadric; minimum distance to the origin

Exact statement. Let $\Sigma \subset \mathbb{R}^3$ be given by

$$2x^2 + 2xy + 4yz + z^2 = 1.$$

By considering a suitable real symmetric matrix, show that there is a new orthonormal basis with associated coordinates u, v, w such that

$$\Sigma : \lambda u^2 + \mu v^2 + \nu w^2 = 1,$$

for constants λ, μ, ν to be determined. Find the minimum distance from a point on Σ to the origin.

Solution. Write the quadratic form as $(x, y, z)M(x, y, z)^T$ with

$$M = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 1 \end{pmatrix},$$

which is real symmetric. By the spectral theorem there is an orthogonal matrix P such that

$$P^T M P = \text{diag}(\lambda, \mu, \nu).$$

The eigenvalues are

$$\lambda = 3, \quad \mu = \sqrt{3}, \quad \nu = -\sqrt{3},$$

so in orthonormal coordinates (u, v, w) one has

$$3u^2 + \sqrt{3}v^2 - \sqrt{3}w^2 = 1.$$

To minimize the distance to the origin, minimize $u^2 + v^2 + w^2$ subject to $3u^2 + \sqrt{3}v^2 - \sqrt{3}w^2 = 1$. Using Lagrange multipliers, the minimum occurs for $v = w = 0$ and $3u^2 = 1$, hence $u^2 = 1/3$ and

$$d_{\min} = \sqrt{u^2} = \frac{1}{\sqrt{3}}.$$

(Indeed, the closest points lie on the eigendirection of eigenvalue 3.)

Problem 12 — Cayley transform of an anti-Hermitian matrix

Exact statement. If S is a real symmetric matrix and A is a real antisymmetric matrix, show that $A + iS$ is anti-hermitian and deduce that

$$\det(A + iS - I) \neq 0.$$

Show that

$$U = (I + A + iS)(I - A - iS)^{-1}$$

is unitary. Find U when

$$S = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and show that it has eigenvalues $\pm(1 - i)/\sqrt{2}$.

Solution. Let $K = A + iS$. Since $A^T = -A$ and $S^T = S$,

$$K^\dagger = (A + iS)^\dagger = A^T - iS^T = -A - iS = -K,$$

so K is anti-hermitian. Hence all eigenvalues of K are purely imaginary. Therefore eigenvalues of $K - I$ are of the form $\mu - 1$ with $\mu \in i\mathbb{R}$, which can never be 0; thus $K - I$ is invertible and $\det(K - I) \neq 0$.

Define

$$U = (I + K)(I - K)^{-1}.$$

Using $K^\dagger = -K$,

$$U^\dagger = ((I - K)^{-1})^\dagger (I + K)^\dagger = (I - K^\dagger)^{-1} (I + K^\dagger) = (I + K)^{-1} (I - K),$$

and hence

$$U^\dagger U = (I + K)^{-1} (I - K) (I + K) (I - K)^{-1} = I,$$

so U is unitary.

For

$$S = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

a direct computation gives

$$U = \begin{pmatrix} 0 & i \\ -1 & 0 \end{pmatrix}.$$

Its characteristic polynomial is

$$\det(U - \lambda I) = \det \begin{pmatrix} -\lambda & i \\ -1 & -\lambda \end{pmatrix} = \lambda^2 + i,$$

so $\lambda^2 = -i$, and therefore

$$\lambda = \pm\sqrt{-i} = \pm e^{-i\pi/4} = \pm \frac{1-i}{\sqrt{2}}.$$

Theory (Problems 1–12): the toolbox

Core objects

- **Eigenvalues/eigenvectors:** λ is an eigenvalue of A iff $\det(A - \lambda I) = 0$.
- **Characteristic polynomial:** $p_A(t) = \det(tI - A)$.
- **Diagonalisation:** $A = PDP^{-1}$, or $P^T AP = D$ when A is real symmetric (orthogonal P).
- **Algebraic vs geometric multiplicity:** $m_{\text{alg}}(\lambda)$ is multiplicity as root of p_A ; $m_{\text{geo}}(\lambda) = \dim \ker(A - \lambda I)$. A is diagonalizable iff $m_{\text{geo}}(\lambda) = m_{\text{alg}}(\lambda)$ for all λ (equivalently: minimal polynomial has no repeated root).

Problem 1: triangular matrices

If A is upper (or lower) triangular, then $A - tI$ is triangular and

$$\det(A - tI) = \prod_{i=1}^n (A_{ii} - t),$$

so the eigenvalues are the diagonal entries.

Problem 2: repeated eigenvalue and Jordan obstruction

Even if $p_A(t) = (t - \lambda)^n$, diagonalizability requires $\dim \ker(A - \lambda I) = n$. If $\dim \ker(A - \lambda I) < n$ then a nontrivial Jordan block occurs and A is not diagonalizable.

Problem 3: orthogonal matrices

Q is orthogonal iff $Q^T Q = I$ iff its columns form an orthonormal basis. In $\mathbb{R}^3$, if v_1, v_2 are orthonormal, then $v_3 = \pm(v_1 \times v_2)$ completes an orthonormal basis.

Problem 4: spectral theorem (symmetric matrices)

If $A = A^T$ then eigenvalues are real, eigenvectors for distinct eigenvalues are orthogonal, and

$$P^T A P = D \quad (P \text{ orthogonal, } D \text{ diagonal}).$$

If D is invertible then $A^{-1} = P D^{-1} P^T$.

Problem 5: quadratic forms and principal axes

Any quadratic form is $q(x) = x^T A x$ with A symmetric. Under an orthogonal change $x = P y$,

$$x^T A x = y^T (P^T A P) y,$$

so diagonalising A removes cross-terms and yields principal axes. Rotations are the 2×2 case.

Problem 6: anti-Hermitian and unitary

If $A^\dagger = -A$, then eigenvalues lie in $i\mathbb{R}$ and eigenvectors for distinct eigenvalues are orthogonal. If $U^\dagger U = I$, then $|\lambda| = 1$ for every eigenvalue λ , and distinct-eigenvalue eigenvectors are orthogonal.

Problem 7: Cayley–Hamilton and recurrences

Cayley–Hamilton: $p_A(A) = 0$. For 2×2 ,

$$A^2 - (\operatorname{tr} A)A + (\det A)I = 0,$$

so every A^n reduces to a linear combination of A and I (recurrence for powers). Repeated roots correspond to Jordan/nilpotent parts (e.g. $A = \lambda I + N$ with $N^2 = 0$).

Problem 8: matrices in general bases

With domain basis E and codomain basis F ,

$$[T(v)]_F = A [v]_E.$$

Under basis changes $E \rightarrow E'$ and $F \rightarrow F'$,

$$A_{F',E'} = C_F^{-1} A_{F,E} C_E.$$

If $n = m$ and the same change is used on both sides: $A' = R^{-1} A R$ (similarity).

Problem 9: similarity to a Jordan block

If $A = P B P^{-1}$ then $A^n = P B^n P^{-1}$. If $B = I + N$ with $N^2 = 0$, then

$$B^n = (I + N)^n = I + nN \quad \Rightarrow \quad B^n - I = n(B - I).$$

Also $\det(A^n) = (\det A)^n$ for every square matrix.

Problem 10: quadric classification via eigenvalues

For symmetric A , the surface $x^T A x = 1$ becomes $\sum \lambda_i u_i^2 = 1$ in an orthonormal eigenbasis. Signs of λ_i classify the quadric (ellipsoid/cylinder/hyperboloid). Positive axis length is $1/\sqrt{\lambda_i}$.

Problem 11: minimum distance to the origin

Diagonalize to $\sum \lambda_i u_i^2 = 1$ and minimize $u_1^2 + u_2^2 + u_3^2$. If A is positive definite, the minimum radius is

$$r_{\min} = \frac{1}{\sqrt{\lambda_{\max}}}$$

(attained along the eigenvector of $\lambda_{\max}$). Indefinite cases are handled similarly by setting negative-direction coordinates to zero when feasible.

Problem 12: Cayley transform

If $K^\dagger = -K$ then $I - K$ is invertible, and

$$U = (I + K)(I - K)^{-1}$$

is unitary. Eigenvalues map by $it \mapsto \frac{1+it}{1-it}$, sending the imaginary axis to the unit circle.

Quick theory table (Problems 1–12)

No.	Core idea / theorem	Key formula(s)	What to check fast
1	Triangular determinant & spectrum	If A triangular, $p_A(t) = \det(A - tI) = \prod_{i=1}^n (A_{ii} - t)$	Eigenvalues are diagonal entries (with multiplicity).
2	Diagonalizable $\Leftrightarrow$ enough eigenvectors (Jordan obstruction)	$m_{\text{geo}}(\lambda) = \dim \ker(A - \lambda I) \leq m_{\text{alg}}(\lambda)$; diag iff equality for all λ	If one eigenvalue λ of mult. n , compute $\dim \ker(A - \lambda I)$; if $< n$, not diagonalizable.
3	Orthogonal matrices and completing an ONB	$Q^T Q = I \iff$ columns orthonormal; in $\mathbb{R}^3$: $v_3 = \pm(v_1 \times v_2)$	Two completions only ($\det = \pm 1$).
4	Spectral theorem (real symmetric)	$A = A^T \Rightarrow P^T A P = D$, P orthogonal; $A^{-1} = P D^{-1} P^T$ if $\lambda_i \neq 0$	Eigenvalues real; eigenvectors for distinct eigenvalues orthogonal.
5	Quadratic forms: principal axes via diagonalization	$q(x) = x^T A x$ (take A symmetric); $x = P y \Rightarrow q = y^T (P^T A P) y$	Remove cross terms by orthogonal change; semi-axes $1/\sqrt{\lambda_i}$ (positive λ_i).
6	Anti-Hermitian / unitary spectra and orthogonality	$A^\dagger = -A \Rightarrow \lambda \in i\mathbb{R}$; $U^\dagger U = I \Rightarrow \lambda = 1$	Distinct eigenvalues $\Rightarrow$ eigenvectors orthogonal (Hermitian inner product).
7	Cayley–Hamilton (esp. 2×2) and power recurrences	$A^2 - (\text{tr } A)A + (\det A)I = 0$; then $A^n = \alpha_n A + \beta_n I$ with recurrences	Use CH to reduce all powers to $\text{span}\{I, A\}$; repeated root $\Rightarrow$ Jordan/unipotent behavior.
8	Matrix of a linear map in general bases; change-of-basis rule	$[T(v)]_F = A[v]_E$; under $E \rightarrow E'$, $F \rightarrow F'$: $A_{F',E'} = C_F^{-1} A_{F,E} C_E$	Columns are coordinates of $T(e_j)$ in codomain basis.
9	Similarity to a Jordan block; fast powers	If $A = P B P^{-1}$ then $A^n = P B^n P^{-1}$; if $B = I + N$, $N^2 = 0$, then $B^n = I + nN$	If $(B - I)^2 = 0$, then $B^n - I = n(B - I)$; transfer by similarity.
10	Quadrics from eigenvalues (classification)	For symmetric A : $x^T A x = 1 \iff \sum \lambda_i u_i^2 = 1$ in ON eigenbasis	All $\lambda_i > 0$: ellipsoid; one $\lambda_i = 0$: cylinder; mixed signs: hyperboloid (signature test).
11	Min distance to origin on a quadric	Diagonalize and minimize $u_1^2 + u_2^2 + u_3^2$ subject to $\sum \lambda_i u_i^2 = 1$	If $A \succ 0$: $r_{\min} = 1/\sqrt{\lambda_{\max}}$ (achieved on eigenvector of $\lambda_{\max}$).
12	Cayley transform: skew-Hermitian $\rightarrow$ unitary	If $K^\dagger = -K$, then $U = (I + K)(I - K)^{-1}$ is unitary; eigenvalue map: $it \mapsto \frac{1+it}{1-it}$	$I - K$ invertible; unit circle image; invariants preserved by similarity/change of basis.

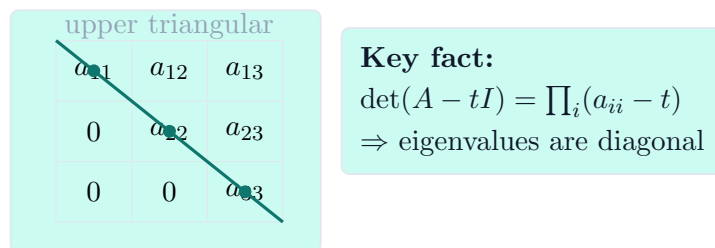


Figure 1: Upper triangular matrices: characteristic polynomial factors along the diagonal.

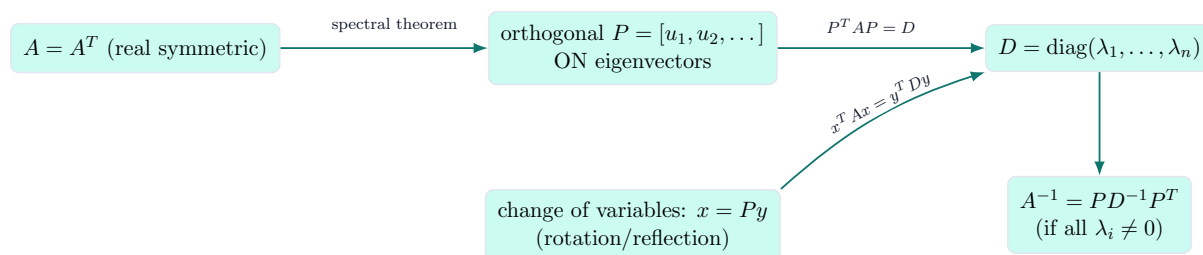
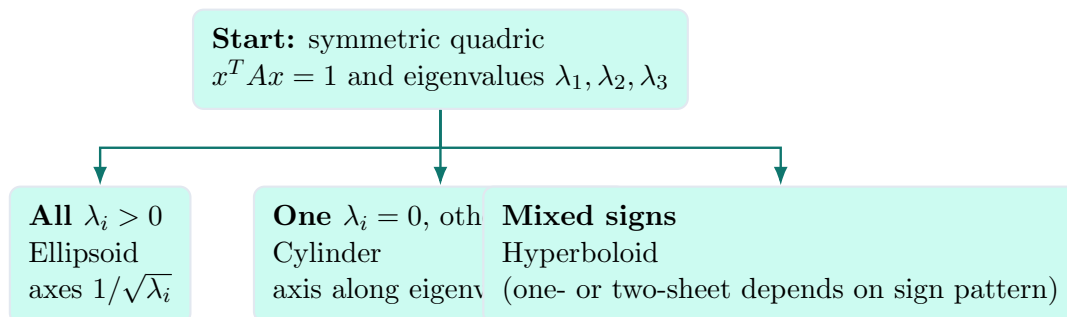


Figure 2: Spectral theorem as an algorithm.



Fast check: signature = #positive, #zero, #negative

Figure 3: Classification of $x^T A x = 1$ by eigenvalue signs (signature).

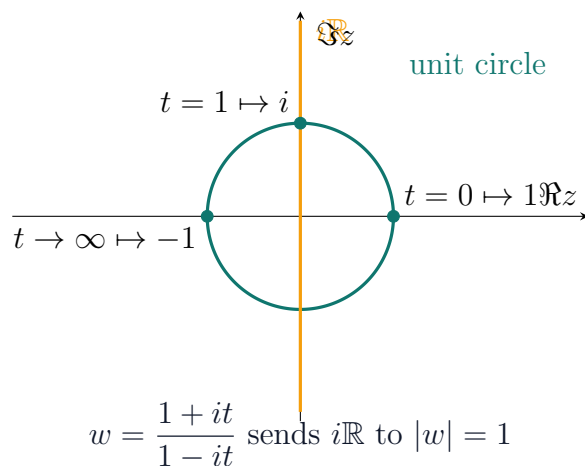
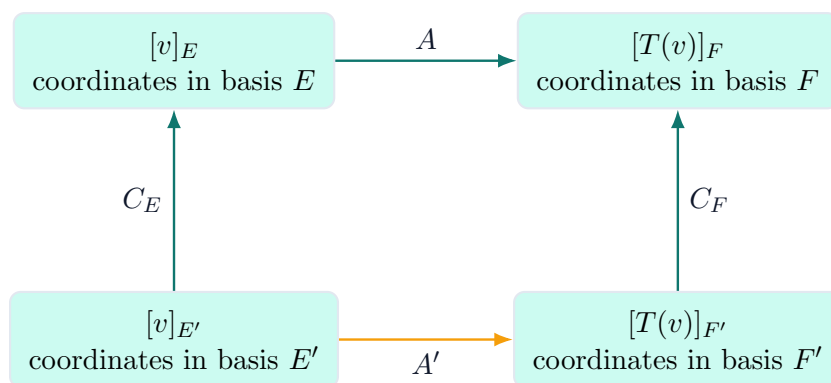


Figure 4: Cayley transform: eigenvalues of skew-Hermitian matrices (imaginary axis) map to unit-modulus eigenvalues (unit circle).



$$A' = C_F^{-1} A C_E$$

(if same basis change both sides: $A' = R^{-1}AR$)

Figure 5: Change-of-basis rule: new matrix is obtained by conjugation with coordinate-change matrices.

Eigenvalues, eigenvectors, orthogonality: methods and worked examples

1. Definitions and the basic workflow

Let $A \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$).

Eigenvalue/eigenvector. A scalar λ is an eigenvalue of A if there exists $v \neq 0$ with

$$Av = \lambda v.$$

Such a vector v is an eigenvector for λ .

Characteristic polynomial.

$$p_A(t) = \det(tI - A).$$

Then λ is an eigenvalue iff $p_A(\lambda) = 0$.

Eigenspace and multiplicities. The eigenspace of λ is

$$E_\lambda = \ker(A - \lambda I).$$

Algebraic multiplicity $m_{\text{alg}}(\lambda)$ is the multiplicity of λ as a root of $p_A(t)$. Geometric multiplicity $m_{\text{geo}}(\lambda) = \dim E_\lambda$.

Diagonalizability. A is diagonalizable iff it has n linearly independent eigenvectors, equivalently

$$\sum_{\lambda} \dim E_\lambda = n,$$

equivalently $m_{\text{geo}}(\lambda) = m_{\text{alg}}(\lambda)$ for every eigenvalue λ .

Standard workflow (practical).

1. Compute $p_A(t) = \det(tI - A)$ and factor it.
2. For each eigenvalue λ , solve $(A - \lambda I)v = 0$ to get eigenvectors (a basis of E_λ).
3. Check diagonalizability by counting eigenvectors: total dimension $\sum_{\lambda} \dim E_\lambda$.
4. If A is symmetric (real): orthogonality is automatic for distinct eigenvalues; then build an orthonormal eigenbasis and use $P^T A P = D$.
5. If needed: normalize eigenvectors, and form $P = [u_1 \ \cdots \ u_n]$.

2. Orthogonality principles (why eigenvectors become perpendicular)

Real symmetric matrices. If $A = A^T$ and $Av = \lambda v$, $Aw = \mu w$ with $\lambda \neq \mu$, then

$$v^T w = 0.$$

Proof sketch: $v^T A w = \mu v^T w$, but also $v^T A w = (A^T v)^T w = (Av)^T w = \lambda v^T w$; hence $(\mu - \lambda)v^T w = 0$.

Anti-Hermitian matrices. If $A^\dagger = -A$, then eigenvalues lie in $i\mathbb{R}$ and eigenvectors for distinct eigenvalues are orthogonal.

Unitary matrices. If $U^\dagger U = I$, then $|\lambda| = 1$ for eigenvalues and distinct-eigenvalue eigenvectors are orthogonal.

3. Example A: triangular matrix (fast eigenvalues)

Let

$$T = \begin{pmatrix} 2 & 5 & -1 \\ 0 & 3 & 4 \\ 0 & 0 & -2 \end{pmatrix}.$$

Step 1: eigenvalues. $T - tI$ is triangular, so

$$p_T(t) = \det(T - tI) = (2 - t)(3 - t)(-2 - t).$$

Hence eigenvalues are $\lambda = 2, 3, -2$.

Step 2: eigenvectors. Solve $(T - \lambda I)v = 0$.

For $\lambda = 2$:

$$T - 2I = \begin{pmatrix} 0 & 5 & -1 \\ 0 & 1 & 4 \\ 0 & 0 & -4 \end{pmatrix}.$$

From bottom: $-4v_3 = 0 \Rightarrow v_3 = 0$. Then $v_2 + 4v_3 = v_2 = 0$. So v_1 free. Take $v = (1, 0, 0)^T$.

For $\lambda = 3$:

$$T - 3I = \begin{pmatrix} -1 & 5 & -1 \\ 0 & 0 & 4 \\ 0 & 0 & -5 \end{pmatrix}.$$

From $4v_3 = 0 \Rightarrow v_3 = 0$. Then $-v_1 + 5v_2 = 0 \Rightarrow v_1 = 5v_2$. Take $v = (5, 1, 0)^T$.

For $\lambda = -2$:

$$T + 2I = \begin{pmatrix} 4 & 5 & -1 \\ 0 & 5 & 4 \\ 0 & 0 & 0 \end{pmatrix}.$$

Let $v_3 = s$ free. From second row: $5v_2 + 4s = 0 \Rightarrow v_2 = -\frac{4}{5}s$. From first: $4v_1 + 5v_2 - s = 0 \Rightarrow 4v_1 + 5(-\frac{4}{5}s) - s = 0 \Rightarrow 4v_1 - 5s = 0 \Rightarrow v_1 = \frac{5}{4}s$. Take $s = 4$ to clear denominators: $v = (5, -16, 4)^T$.

Conclusion. For distinct diagonal entries, triangular matrices are diagonalizable.

4. Example B: symmetric matrix (orthonormal eigenbasis)

Let

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}.$$

Step 1: eigenvalues. Compute $p_A(t) = \det(tI - A)$ and factor (or use a CAS; here it factors as)

$$p_A(t) = (t - 1)(t - 2)(t - 4).$$

So $\lambda = 1, 2, 4$.

Step 2: eigenvectors (solve each system).

- For $\lambda = 1$, a solution is $v_1 = (-1, 1, 1)^T$.

- For $\lambda = 2$, a solution is $v_2 = (0, -1, 1)^T$.
- For $\lambda = 4$, a solution is $v_3 = (2, 1, 1)^T$.

Step 3: orthogonality and normalization. Because A is symmetric, these eigenvectors are automatically orthogonal (check quickly):

$$v_1 \cdot v_2 = 0, \quad v_1 \cdot v_3 = 0, \quad v_2 \cdot v_3 = 0.$$

Normalize:

$$u_1 = \frac{1}{\sqrt{3}}(-1, 1, 1), \quad u_2 = \frac{1}{\sqrt{2}}(0, -1, 1), \quad u_3 = \frac{1}{\sqrt{6}}(2, 1, 1).$$

Step 4: diagonalization and inverse. Let $P = [u_1 \ u_2 \ u_3]$, $D = \text{diag}(1, 2, 4)$. Then

$$P^T A P = D, \quad A^{-1} = P D^{-1} P^T.$$

5. Example C: repeated eigenvalue but not diagonalizable (Jordan obstruction)

Let

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Step 1: eigenvalues.

$$p_J(t) = \det(tI - J) = \det \begin{pmatrix} t-1 & -1 \\ 0 & t-1 \end{pmatrix} = (t-1)^2.$$

Only eigenvalue is $\lambda = 1$ with $m_{\text{alg}} = 2$.

Step 2: eigenspace.

$$J - I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \Rightarrow (J - I)v = 0 \Rightarrow v_2 = 0.$$

So $E_1 = \{(v_1, 0)\}$ has dimension 1.

Conclusion. $m_{\text{geo}}(1) = 1 < 2 = m_{\text{alg}}(1)$, so J is **not** diagonalizable.

Bonus: fast powers. Write $J = I + N$ with $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $N^2 = 0$. Then

$$J^n = (I + N)^n = I + nN = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}.$$

6. Example D: unitary / anti-Hermitian quick spectral facts

Anti-Hermitian example. Let

$$K = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

Then $K^\dagger = -K$, so eigenvalues are purely imaginary. Indeed,

$$p_K(t) = \det(tI - K) = \det \begin{pmatrix} t & -i \\ -i & t \end{pmatrix} = t^2 + 1,$$

so $\lambda = \pm i$. Distinct eigenvalues $\Rightarrow$ eigenvectors orthogonal.

Unitary example. Let

$$U = \begin{pmatrix} 0 & i \\ -1 & 0 \end{pmatrix}.$$

One checks $U^\dagger U = I$. Its eigenvalues satisfy $|\lambda| = 1$; indeed,

$$\det(U - \lambda I) = \lambda^2 + i = 0 \Rightarrow \lambda = \pm \frac{1-i}{\sqrt{2}},$$

which lie on the unit circle.

7. Example E: quadratic forms and principal axes (3D geometry)

Given a symmetric quadratic form $q(x) = x^T A x$, the surface $q(x) = 1$ becomes

$$\lambda_1 u_1^2 + \lambda_2 u_2^2 + \lambda_3 u_3^2 = 1$$

in an orthonormal eigenbasis. The signs of λ_i classify the quadric:

- all $\lambda_i > 0$: ellipsoid,
- one $\lambda_i = 0$ and others > 0 : cylinder,
- mixed signs: hyperboloid.

If $\lambda_i > 0$, the semi-axis length along eigenvector u_i is $1/\sqrt{\lambda_i}$.

Distance minimization trick. For $x^T A x = 1$ with $A \succ 0$ (positive definite), the nearest point to the origin lies on the eigenvector for $\lambda_{\max}$, and

$$r_{\min} = \frac{1}{\sqrt{\lambda_{\max}}}.$$

This comes from minimizing $\|x\|^2$ under the constraint $\sum \lambda_i u_i^2 = 1$.

Problem 1 — Basic identities for Hermitian matrices and unitary similarity

Exact statement. Let A and B be $n \times n$ Hermitian matrices and U be an $n \times n$ unitary matrix. Show that (i) $\text{tr } A$ and $\det A$ are real; (ii) $AB + BA$ is Hermitian; (iii) $i(AB - BA)$ is Hermitian and has zero trace; (iv) $\text{tr}(AB)$ and $\det(AB)$ are real; (v) $\text{tr}(UAU^*) = \text{tr}(A)$ and $\det(UAU^*) = \det(A)$.

Solution. Recall: Hermitian means $A^* = A$, and unitary means $U^*U = I$. Also $\overline{\text{tr}(X)} = \text{tr}(X^*)$, $\overline{\det(X)} = \det(X^*)$, and $\text{tr}(XY) = \text{tr}(YX)$.

(i) Since $A^* = A$,

$$\overline{\operatorname{tr}(A)} = \operatorname{tr}(A^*) = \operatorname{tr}(A),$$

so $\operatorname{tr}(A) \in \mathbb{R}$. Also

$$\overline{\det(A)} = \det(A^*) = \det(A),$$

so $\det(A) \in \mathbb{R}$.

(ii)

$$(AB + BA)^* = B^*A^* + A^*B^* = BA + AB = AB + BA.$$

(iii)

$$(AB - BA)^* = B^*A^* - A^*B^* = BA - AB = -(AB - BA),$$

so $AB - BA$ is skew-Hermitian, hence

$$(i(AB - BA))^* = -i(AB - BA)^* = i(AB - BA),$$

so $i(AB - BA)$ is Hermitian. Moreover,

$$\operatorname{tr}(AB - BA) = \operatorname{tr}(AB) - \operatorname{tr}(BA) = 0.$$

(iv)

$$\overline{\operatorname{tr}(AB)} = \operatorname{tr}((AB)^*) = \operatorname{tr}(B^*A^*) = \operatorname{tr}(BA) = \operatorname{tr}(AB),$$

so $\operatorname{tr}(AB) \in \mathbb{R}$. Also $\det(AB) = \det(A)\det(B)$, and by (i) both determinants are real, hence $\det(AB) \in \mathbb{R}$.

(v) Using cyclicity,

$$\operatorname{tr}(UAU^*) = \operatorname{tr}(AU^*U) = \operatorname{tr}(A).$$

Also

$$\det(UAU^*) = \det(U)\det(A)\det(U^*) = \det(U)\det(A)\overline{\det(U)}.$$

For unitary U , $|\det(U)| = 1$, so $\det(U)\overline{\det(U)} = 1$, hence $\det(UAU^*) = \det(A)$.

Problem 2 — Determinant factorization by row operations

Exact statement. For the real matrix

$$M = \begin{pmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{pmatrix},$$

show (using row operations) that

$$\det M = (a - b)(b - c)(c - a)(ab + bc + ca).$$

Solution. Let the rows be R_1, R_2, R_3 . Perform

$$R_2 \leftarrow R_2 - R_1, \quad R_3 \leftarrow R_3 - R_1.$$

Then

$$\begin{aligned} R_2 - R_1 &= (b - a, b^2 - a^2, ca - bc) = (b - a)(1, a + b, -c), \\ R_3 - R_1 &= (c - a, c^2 - a^2, ab - bc) = (c - a)(1, a + c, -b). \end{aligned}$$

Hence

$$\det M = (b - a)(c - a) \det \begin{pmatrix} a & a^2 & bc \\ 1 & a + b & -c \\ 1 & a + c & -b \end{pmatrix}.$$

Now do $R_3 \leftarrow R_3 - R_2$:

$$(1, a + c, -b) - (1, a + b, -c) = (0, c - b, c - b) = (c - b)(0, 1, 1),$$

so

$$\det M = (b - a)(c - a)(c - b) \det \begin{pmatrix} a & a^2 & bc \\ 1 & a + b & -c \\ 0 & 1 & 1 \end{pmatrix}.$$

Compute the remaining determinant by expansion along the first row:

$$\begin{aligned} \det \begin{pmatrix} a & a^2 & bc \\ 1 & a + b & -c \\ 0 & 1 & 1 \end{pmatrix} &= a((a + b) \cdot 1 - (-c) \cdot 1) - a^2 \cdot 1 + bc \cdot 1 \\ &= a(a + b + c) - a^2 + bc = a(b + c) + bc = ab + ac + bc. \end{aligned}$$

Therefore

$$\det M = (b - a)(c - a)(c - b)(ab + bc + ca) = (a - b)(b - c)(c - a)(ab + bc + ca).$$

Problem 3 — A circulant determinant and its factorization

Exact statement. Show by direct evaluation that

$$\Delta(x, y, z) \equiv \begin{vmatrix} x & y & z \\ z & x & y \\ y & z & x \end{vmatrix} = x^3 + y^3 + z^3 - 3xyz.$$

Now show that $x + y + z$, $x + \omega y + \omega^2 z$, $x + \omega^2 y + \omega z$ are factors of Δ , where $\omega^3 = 1$, $\omega \neq 1$.

Solution. Direct evaluation. Expanding along the first row,

$$\Delta = x \begin{vmatrix} x & y \\ z & x \end{vmatrix} - y \begin{vmatrix} z & y \\ y & x \end{vmatrix} + z \begin{vmatrix} z & x \\ y & z \end{vmatrix} = x(x^2 - yz) - y(zx - y^2) + z(z^2 - xy),$$

so

$$\Delta = x^3 + y^3 + z^3 - 3xyz.$$

Factors via eigenvectors. Let

$$C = \begin{pmatrix} x & y & z \\ z & x & y \\ y & z & x \end{pmatrix}.$$

Using $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$, check:

$$C \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = (x + y + z) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad C \begin{pmatrix} 1 \\ \omega \\ \omega^2 \end{pmatrix} = (x + \omega y + \omega^2 z) \begin{pmatrix} 1 \\ \omega \\ \omega^2 \end{pmatrix},$$

$$C \begin{pmatrix} 1 \\ \omega^2 \\ \omega \end{pmatrix} = (x + \omega^2 y + \omega z) \begin{pmatrix} 1 \\ \omega^2 \\ \omega \end{pmatrix}.$$

Hence $\det(C) = 0$ whenever any of the three linear factors vanishes, so each factor divides Δ . Since Δ is homogeneous of degree 3, we must have

$$\Delta = c(x + y + z)(x + \omega y + \omega^2 z)(x + \omega^2 y + \omega z)$$

for a constant c . Setting $y = z = 0$ gives $\Delta(x, 0, 0) = x^3$ and the product equals x^3 , so $c = 1$.

Problem 4 — Determinant of an odd-dimensional antisymmetric matrix

Exact statement. If A is a $(2n + 1) \times (2n + 1)$ antisymmetric matrix, find $\det A$.

Solution. Antisymmetric means $A^T = -A$. Then

$$\det(A) = \det(A^T) = \det(-A) = (-1)^{2n+1} \det(A) = -\det(A),$$

so $\det(A) = 0$.

Problem 5 — Determinant of a “ p on diagonal, 1 elsewhere” matrix

Exact statement. Let D be the $n \times n$ matrix ($n > 1$) which has entries p on the main diagonal and entries 1 elsewhere. Show that

$$\det D = (p + n - 1)(p - 1)^{n-1}.$$

Solution. Write

$$D = (p - 1)I + J,$$

where J is the all-ones matrix. The eigenvalues of J are n (with eigenvector $\mathbf{1} = (1, \dots, 1)^T$) and 0 with multiplicity $n - 1$. Hence the eigenvalues of D are $p + n - 1$ (once) and $p - 1$ (multiplicity $n - 1$). Therefore,

$$\det D = (p + n - 1)(p - 1)^{n-1}.$$

Problem 6 — Cofactors, inverse, and solving a linear system

Exact statement. Calculate the cofactors Δ_{ij} for

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & -2 & 2 \end{pmatrix}.$$

Convert $A_{ij}\Delta_{ik} = \delta_{jk} \det A$ into matrix notation and check it. Find A^{-1} . Use it to solve

$$\begin{cases} x + y + z = 1 \\ x + 2y + 3z = -5 \\ 3x - 2y + 2z = 4. \end{cases}$$

Solution. The cofactor matrix $C = (\Delta_{ij})$ is

$$C = \begin{pmatrix} 10 & 7 & -8 \\ -4 & -1 & 5 \\ 1 & -2 & 1 \end{pmatrix}, \quad \det(A) = 9.$$

The suffix identity $A_{ij}\Delta_{ik} = \delta_{jk} \det(A)$ (sum over i) is exactly the matrix identity

$$A^T C = \det(A) I.$$

Equivalently, with $\text{adj}(A) = C^T$,

$$A \text{adj}(A) = \det(A) I.$$

Thus

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A) = \frac{1}{9} C^T = \frac{1}{9} \begin{pmatrix} 10 & -4 & 1 \\ 7 & -1 & -2 \\ -8 & 5 & 1 \end{pmatrix}.$$

To solve $A(x, y, z)^T = (1, -5, 4)^T$,

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} 1 \\ -5 \\ 4 \end{pmatrix} = \begin{pmatrix} \frac{34}{9} \\ \frac{4}{9} \\ -\frac{29}{9} \end{pmatrix}.$$

A direct substitution verifies the system.

Problem 7 — A linear system depending on a parameter t

Exact statement. For each real value of t , determine whether or not there exist solutions to the simultaneous equations

$$\begin{cases} x + y + z = t, \\ tx + 2z = 3, \\ 3x + ty + 5z = 7, \end{cases}$$

exhibiting the most general form of such solutions when they exist.

Solution. Let

$$A(t) = \begin{pmatrix} 1 & 1 & 1 \\ t & 0 & 2 \\ 3 & t & 5 \end{pmatrix}, \quad b(t) = \begin{pmatrix} t \\ 3 \\ 7 \end{pmatrix}.$$

Then

$$\det A(t) = \begin{vmatrix} 1 & 1 & 1 \\ t & 0 & 2 \\ 3 & t & 5 \end{vmatrix} = t^2 - 7t + 6 = (t-1)(t-6).$$

Case 1: $t \neq 1, 6$. Then $\det A(t) \neq 0$, so there is a unique solution. Solving (e.g. by elimination) gives

$$x = \frac{1-2t}{t-6}, \quad y = \frac{8-5t}{t-6}, \quad z = \frac{t^2+t-9}{t-6}.$$

Case 2: $t = 1$. The system becomes

$$\begin{cases} x + y + z = 1, \\ x + 2z = 3, \\ 3x + y + 5z = 7. \end{cases}$$

From $x + 2z = 3$ we get $x = 3 - 2z$. Substituting into $3x + y + 5z = 7$ yields

$$3(3 - 2z) + y + 5z = 7 \Rightarrow y - z = -2 \Rightarrow y = z - 2.$$

Thus, writing $z = s$,

$$(x, y, z) = (3 - 2s, s - 2, s), \quad s \in \mathbb{R},$$

equivalently $(x, y, z) = (3, -2, 0) + s(-2, 1, 1)$.

Case 3: $t = 6$. The system becomes

$$\begin{cases} x + y + z = 6, \\ 6x + 2z = 3, \\ 3x + 6y + 5z = 7. \end{cases}$$

From $6x + 2z = 3$ we get $3x + z = \frac{3}{2}$. Also,

$$(3x + 6y + 5z) - 3(x + y + z) = 7 - 18 \Rightarrow 3y + 2z = -11.$$

Using $y = 6 - x - z$ gives

$$3(6 - x - z) + 2z = -11 \Rightarrow 18 - 3x - z = -11 \Rightarrow 3x + z = 29,$$

contradicting $3x + z = \frac{3}{2}$. Hence there are no solutions when $t = 6$.

Problem 8 — Image, kernel, and solvability of $Mx = d$ (Vandermonde-type matrix)

Exact statement. Let M be a real 3×3 matrix, and let d be a 3 component column vector. Explain briefly how the general solution of the matrix equation $Mx = d$, where x is a 3 component column vector, depends on the image and kernel of the linear map $x \mapsto Mx$.

Consider the case

$$M = \begin{pmatrix} 1 & 1 & 1 \\ 1 & a & b \\ 1 & a^2 & b^2 \end{pmatrix}, \quad x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad d = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}.$$

Find the image and kernel of the corresponding map, noting the different possibilities according to different values of a and b . For which values of a and b do the equations $Mx = d$ have (i) a unique solution, (ii) more than one solution, (iii) no solution? For each pair (a, b) satisfying (ii), give the solutions as the sum of a fixed solution and the general solution of the corresponding homogeneous equations.

Solution. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be $T(x) = Mx$. Then:

$$Mx = d \text{ is solvable} \iff d \in \text{Im}(M),$$

and if x_0 is one solution, then all solutions are

$$x = x_0 + v, \quad v \in \ker(M).$$

The solution is unique iff $\ker(M) = \{0\}$, equivalently $\det(M) \neq 0$.

Write the columns of M as

$$c_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad c_2 = \begin{pmatrix} 1 \\ a \\ a^2 \end{pmatrix}, \quad c_3 = \begin{pmatrix} 1 \\ b \\ b^2 \end{pmatrix}.$$

Then $\text{Im}(M) = \text{span}\{c_1, c_2, c_3\}$ and a direct computation gives

$$\det M = (a - 1)(b - 1)(b - a).$$

Image and kernel (by cases).

(A) If $(a - 1)(b - 1)(a - b) \neq 0$. Then M is invertible, so

$$\ker(M) = \{0\}, \quad \text{Im}(M) = \mathbb{R}^3.$$

(B) Rank 2 cases (exactly two of 1, a , b coincide).

- If $a = 1$ and $b \neq 1$, then $c_2 = c_1$, hence

$$\ker(M) = \text{span}\{(1, -1, 0)\}, \quad \text{Im}(M) = \text{span}\left\{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ b \\ b^2 \end{pmatrix}\right\}.$$

- If $b = 1$ and $a \neq 1$, then $c_3 = c_1$, hence

$$\ker(M) = \text{span}\{(1, 0, -1)\}, \quad \text{Im}(M) = \text{span}\left\{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ a \\ a^2 \end{pmatrix}\right\}.$$

- If $a = b \neq 1$, then $c_2 = c_3$, hence

$$\ker(M) = \text{span}\{(0, 1, -1)\}, \quad \text{Im}(M) = \text{span}\left\{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ a \\ a^2 \end{pmatrix}\right\}.$$

(C) Rank 1 case. If $a = b = 1$, then $c_1 = c_2 = c_3$, hence

$$\text{Im}(M) = \text{span}\left\{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\}, \quad \ker(M) = \{(u, v, w) \in \mathbb{R}^3 : u + v + w = 0\}.$$

Solving $Mx = d$ for $d = (1, -1, 1)^T$.

(i) *Unique solution.* This occurs exactly when $\det M \neq 0$, i.e. when $(a-1)(b-1)(a-b) \neq 0$. Then

$$x = \frac{(a+1)(b+1)}{(a-1)(b-1)}, \quad y = \frac{2(b+1)}{(a-1)(a-b)}, \quad z = -\frac{2(a+1)}{(b-1)(a-b)}.$$

(ii) *More than one solution.* This requires $\det M = 0$ and $d \in \text{Im}(M)$. This happens only for

$$(a, b) = (1, -1), \quad (a, b) = (-1, 1), \quad (a, b) = (-1, -1).$$

For these pairs:

- If $(a, b) = (1, -1)$, one fixed solution is $(0, 0, 1)$, and $\ker(M) = \text{span}\{(1, -1, 0)\}$, so

$$(x, y, z) = (0, 0, 1) + s(1, -1, 0), \quad s \in \mathbb{R}.$$

- If $(a, b) = (-1, 1)$, one fixed solution is $(0, 1, 0)$, and $\ker(M) = \text{span}\{(1, 0, -1)\}$, so

$$(x, y, z) = (0, 1, 0) + s(1, 0, -1), \quad s \in \mathbb{R}.$$

- If $(a, b) = (-1, -1)$, one fixed solution is $(0, 1, 0)$, and $\ker(M) = \text{span}\{(0, 1, -1)\}$, so

$$(x, y, z) = (0, 1, 0) + s(0, 1, -1), \quad s \in \mathbb{R}.$$

(iii) *No solution.* For all other (a, b) with $\det M = 0$, one has $d \notin \text{Im}(M)$, hence no solutions exist.

Problem 10 — Eigenvalues/eigenvectors; reality + orthogonality $\Leftrightarrow$ Hermitian

Exact statement. Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 1 & \alpha & 0 \\ \beta & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

where α, β are non-zero complex numbers. Find the conditions for which (i) the eigenvalues are real and (ii) eigenvectors for different eigenvalues are orthogonal. Show that both these conditions hold if and only if A is hermitian.

Recall: the complex inner product on $\mathbb{C}^3$ is

$$z^\dagger w = \bar{z}_1 w_1 + \bar{z}_2 w_2 + \bar{z}_3 w_3,$$

and z, w are orthogonal if $z^\dagger w = 0$.

Solution. Write $A = \text{diag}(B, 1)$ with

$$B = \begin{pmatrix} 1 & \alpha \\ \beta & 1 \end{pmatrix}.$$

Then

$$\chi_B(\lambda) = \det(B - \lambda I) = (1 - \lambda)^2 - \alpha\beta,$$

so with $s = \sqrt{\alpha\beta}$ we have eigenvalues

$$\lambda_1 = 1 + s, \quad \lambda_2 = 1 - s, \quad \lambda_3 = 1.$$

For $\lambda = 1 + s$, solving $(B - (1 + s)I)v = 0$ gives $-su + \alpha v = 0$, so we may take

$$v_+ = \begin{pmatrix} \alpha \\ s \\ 0 \end{pmatrix}.$$

For $\lambda = 1 - s$, we may take

$$v_- = \begin{pmatrix} \alpha \\ -s \\ 0 \end{pmatrix}.$$

For $\lambda = 1$, an eigenvector is $v_0 = e_3 = (0, 0, 1)^T$.

(i) Real eigenvalues. The eigenvalues are real iff $s \in \mathbb{R}$, i.e. iff $\alpha\beta \in (0, \infty) \subset \mathbb{R}$.

(ii) Orthogonality for distinct eigenvalues. Clearly $v_0 \perp v_\pm$. Thus we require $v_+^\dagger v_- = 0$:

$$v_+^\dagger v_- = \bar{\alpha}\alpha + \bar{s}(-s) = |\alpha|^2 - |s|^2.$$

If (i) holds then $s \in \mathbb{R}$ so $|s|^2 = s^2 = \alpha\beta$, hence

$$v_+^\dagger v_- = 0 \iff |\alpha|^2 = \alpha\beta \iff \beta = \bar{\alpha} \quad (\alpha \neq 0).$$

But $\beta = \bar{\alpha}$ is exactly the condition that A is hermitian. Conversely, if A is hermitian then $\alpha\beta = \alpha\bar{\alpha} = |\alpha|^2 > 0$ so (i) holds, and eigenvectors for distinct eigenvalues are orthogonal.

Problem 11 — Odd-dimensional orthogonal matrices with $\det = 1$ have eigenvalue 1

Exact statement. Let Q be a $(2n + 1) \times (2n + 1)$ orthogonal matrix with $\det Q = 1$. Show that 1 is an eigenvalue of Q and give a geometric interpretation of this result when Q is a 3×3 matrix. What can be said if $\det Q = -1$?

Solution. If Q is real orthogonal then $Q^T Q = I$, hence every eigenvalue λ satisfies $|\lambda| = 1$. Non-real eigenvalues occur in complex conjugate pairs $\lambda, \bar{\lambda}$, and each pair contributes product $\lambda \bar{\lambda} = |\lambda|^2 = 1$ to $\det Q$. Since the dimension is odd, at least one eigenvalue is real; real eigenvalues of an orthogonal matrix are ± 1 . Thus the sign of $\det Q$ is determined by the number of eigenvalues -1 . If $\det Q = 1$, the number of -1 eigenvalues is even, so among the (odd number of) real eigenvalues there must be at least one $+1$. Hence 1 is an eigenvalue.

For 3×3 , $\det Q = 1$ means Q is a rotation: the eigenvector with eigenvalue 1 is the rotation axis (a fixed direction). If $\det Q = -1$, then -1 is an eigenvalue; geometrically this corresponds to an improper orthogonal map (reflection or roto-reflection).

Problem 12 — Eigenvalues, eigenspace dimensions, diagonalizability

Exact statement. For each of the matrices

$$A = \begin{pmatrix} 5 & -3 & 2 \\ 6 & -4 & 4 \\ 4 & -4 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -3 & 4 \\ 4 & -7 & 8 \\ 6 & -7 & 7 \end{pmatrix}, \quad C = \begin{pmatrix} 7 & -12 & 6 \\ 10 & -19 & 10 \\ 12 & -24 & 13 \end{pmatrix},$$

- (i) compute their eigenvalues; (ii) for each eigenvalue λ compute $\dim\{x \in \mathbb{R}^3 : Mx = \lambda x\}$; (iii) hence determine whether or not the matrix is diagonalizable.

Solution.

Matrix A .

$$\chi_A(\lambda) = \det(\lambda I - A) = \lambda^3 - 6\lambda^2 + 11\lambda - 6 = (\lambda - 1)(\lambda - 2)(\lambda - 3).$$

Hence the eigenvalues are 1, 2, 3, each with algebraic multiplicity 1, so each eigenspace has dimension 1. For example,

$$\lambda = 1 : (1, 2, 1), \quad \lambda = 2 : (1, 1, 0), \quad \lambda = 3 : (1, 2, 2)$$

are eigenvectors. Therefore A is diagonalizable.

Matrix B .

$$\chi_B(\lambda) = \lambda^3 - \lambda^2 - 5\lambda - 3 = (\lambda - 3)(\lambda + 1)^2.$$

So the eigenvalues are 3 and -1 (with algebraic multiplicity 2). One finds

$$E_3 = \ker(B - 3I) = \text{span}\{(1, 2, 2)\}, \quad \dim E_3 = 1,$$

$$E_{-1} = \ker(B + I) = \text{span}\{(1, 2, 1)\}, \quad \dim E_{-1} = 1.$$

Since $\dim E_{-1} = 1 < 2$, the matrix B is not diagonalizable.

Matrix C .

$$\chi_C(\lambda) = \lambda^3 - \lambda^2 - \lambda + 1 = (\lambda - 1)^2(\lambda + 1).$$

So the eigenvalues are 1 (algebraic multiplicity 2) and -1 . One finds

$$E_1 = \ker(C - I) = \text{span}\{(2, 1, 0), (-1, 0, 1)\}, \quad \dim E_1 = 2,$$

$$E_{-1} = \ker(C + I) = \text{span}\{(3, 5, 6)\}, \quad \dim E_{-1} = 1.$$

Thus $\dim E_1 + \dim E_{-1} = 2 + 1 = 3$, so C is diagonalizable.

Problem 13 — Characteristic polynomial of A^{-1}

Exact statement. Suppose that A is an $n \times n$ matrix and that A^{-1} exists. Show that if A has characteristic polynomial

$$\chi_A(t) = a_0 + a_1t + \cdots + a_nt^n,$$

then A^{-1} has characteristic polynomial

$$\chi_{A^{-1}}(t) = (-1)^n \det(A^{-1})(a_n + a_{n-1}t + \cdots + a_0t^n).$$

Solution. Start with

$$\chi_{A^{-1}}(t) = \det(A^{-1} - tI).$$

Factor out A^{-1} :

$$\det(A^{-1} - tI) = \det(A^{-1}) \det(I - tA).$$

Now

$$I - tA = -t\left(A - \frac{1}{t}I\right),$$

hence

$$\det(I - tA) = (-t)^n \det\left(A - \frac{1}{t}I\right) = (-1)^n t^n \chi_A\left(\frac{1}{t}\right).$$

Therefore

$$\chi_{A^{-1}}(t) = (-1)^n \det(A^{-1}) t^n \chi_A\left(\frac{1}{t}\right).$$

If $\chi_A(t) = a_0 + a_1t + \cdots + a_nt^n$, then

$$t^n \chi_A(1/t) = a_n + a_{n-1}t + \cdots + a_0t^n,$$

so

$$\chi_{A^{-1}}(t) = (-1)^n \det(A^{-1})(a_n + a_{n-1}t + \cdots + a_0t^n),$$

as required.

Problem 1 – Index notation with δ_{ij} and ε_{ijk}

Exact statement. In the following, the indices i, j, k, ℓ take the values 1, 2, 3 and the summation convention applies.

- (a) Simplify the following expressions:

$$\delta_{ij}v_j, \quad \delta_{ij}\delta_{jk}, \quad \delta_{ij}\delta_{ji}, \quad \delta_{ij}v_i v_j, \quad \varepsilon_{ijk}\delta_{ik}, \quad \varepsilon_{ijk}v_j v_k, \quad \varepsilon_{ijk}\varepsilon_{ile}, \quad \varepsilon_{ijk}\varepsilon_{ikj}.$$

- (b) Given that $A_{ij} = \varepsilon_{ijk}a_k$ (for all i, j), show that $2a_k = \varepsilon_{kij}A_{ij}$ (for all k).

- (c) Show that $\varepsilon_{ijk}S_{ij} = 0$ (for all k) if and only if $S_{ij} = S_{ji}$ (for all i, j).

Solution.

- (a) Using δ_{ij} to replace indices and antisymmetry of ε_{ijk} :

$$\begin{aligned} \delta_{ij}v_j &= v_i, & \delta_{ij}\delta_{jk} &= \delta_{ik}, & \delta_{ij}\delta_{ji} &= \delta_{ii} = 3, \\ \delta_{ij}v_i v_j &= v_i v_i = \|v\|^2, & \varepsilon_{ijk}\delta_{ik} &= \varepsilon_{iji} = 0, & \varepsilon_{ijk}v_j v_k &= 0, \\ \varepsilon_{ijk}\varepsilon_{ile} &= \delta_{jl}\delta_{ke} - \delta_{je}\delta_{kl}, & \varepsilon_{ijk}\varepsilon_{ikj} &= -\varepsilon_{ijk}\varepsilon_{ijk} = -6. \end{aligned}$$

- (b) Contract with ε_{kij} :

$$\varepsilon_{kij}A_{ij} = \varepsilon_{kij}\varepsilon_{ij\ell}a_\ell = 2\delta_{k\ell}a_\ell = 2a_k.$$

- (c) If $S_{ij} = S_{ji}$ then $\varepsilon_{ijk}S_{ij} = 0$. Conversely, if $\varepsilon_{ijk}S_{ij} = 0$ for all k , multiply by ε_{kpq} :

$$0 = \varepsilon_{kpq}\varepsilon_{ijk}S_{ij} = (\delta_{pi}\delta_{qj} - \delta_{pj}\delta_{qi})S_{ij} = S_{pq} - S_{qp},$$

$$\text{so } S_{pq} = S_{qp}.$$

Problem 2 – Triple products in $\mathbb{R}^3$

Exact statement. For vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$, simplify $\varepsilon_{ijk}(\mathbf{a} \times \mathbf{b})_k$ and deduce a standard formula for $\mathbf{c} \times (\mathbf{a} \times \mathbf{b})$.

- (a) Let $\mathbf{m}, \mathbf{u}, \mathbf{a}$ be fixed vectors in $\mathbb{R}^3$ such that $\mathbf{m} \cdot \mathbf{u} = 0$ and $\mathbf{a} \cdot \mathbf{u} \neq 0$. Show that the line $\mathbf{r} \times \mathbf{u} = \mathbf{m}$ meets the plane $\mathbf{r} \cdot \mathbf{a} = \kappa$ in the point

$$\mathbf{r} = \frac{\mathbf{a} \times \mathbf{m} + \kappa \mathbf{u}}{\mathbf{a} \cdot \mathbf{u}}.$$

Explain clearly the geometrical meaning of the condition $\mathbf{a} \cdot \mathbf{u} \neq 0$.

- (b) Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ with $\mathbf{a} \times \mathbf{b} \neq 0$. Show that the planes $\mathbf{r} \cdot \mathbf{a} = \kappa$ and $\mathbf{r} \cdot \mathbf{b} = \rho$ (where κ, ρ are constants) intersect in the line

$$\mathbf{r} \times (\mathbf{a} \times \mathbf{b}) = \rho \mathbf{a} - \kappa \mathbf{b}.$$

What happens if $\mathbf{a} \times \mathbf{b} = 0$?

Solution. First,

$$(\mathbf{a} \times \mathbf{b})_k = \varepsilon_{klm} a_l b_m \Rightarrow \varepsilon_{ijk} (\mathbf{a} \times \mathbf{b})_k = \varepsilon_{ijk} \varepsilon_{klm} a_l b_m = a_i b_j - a_j b_i.$$

Hence

$$(\mathbf{c} \times (\mathbf{a} \times \mathbf{b}))_i = \varepsilon_{ijk} c_j (\mathbf{a} \times \mathbf{b})_k = a_i (\mathbf{c} \cdot \mathbf{b}) - b_i (\mathbf{c} \cdot \mathbf{a}),$$

so

$$\mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{a}(\mathbf{c} \cdot \mathbf{b}) - \mathbf{b}(\mathbf{c} \cdot \mathbf{a}).$$

(a) Let $\mathbf{r} = (\mathbf{a} \times \mathbf{m} + \kappa \mathbf{u}) / (\mathbf{a} \cdot \mathbf{u})$. Then

$$\mathbf{r} \times \mathbf{u} = \frac{(\mathbf{a} \times \mathbf{m}) \times \mathbf{u}}{\mathbf{a} \cdot \mathbf{u}} = \frac{\mathbf{m}(\mathbf{a} \cdot \mathbf{u}) - \mathbf{a}(\mathbf{m} \cdot \mathbf{u})}{\mathbf{a} \cdot \mathbf{u}} = \mathbf{m}$$

because $\mathbf{m} \cdot \mathbf{u} = 0$, and

$$\mathbf{r} \cdot \mathbf{a} = \frac{(\mathbf{a} \times \mathbf{m}) \cdot \mathbf{a} + \kappa(\mathbf{u} \cdot \mathbf{a})}{\mathbf{a} \cdot \mathbf{u}} = \kappa.$$

Geometrically, $\mathbf{r} \times \mathbf{u} = \mathbf{m}$ is an affine line parallel to $\mathbf{u}$; if $\mathbf{a} \cdot \mathbf{u} = 0$ the line is parallel to the plane $\mathbf{r} \cdot \mathbf{a} = \kappa$, giving no intersection or infinitely many.

(b) Use

$$\mathbf{r} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{a}(\mathbf{r} \cdot \mathbf{b}) - \mathbf{b}(\mathbf{r} \cdot \mathbf{a}).$$

If $\mathbf{r} \cdot \mathbf{a} = \kappa$ and $\mathbf{r} \cdot \mathbf{b} = \rho$, then $\mathbf{r} \times (\mathbf{a} \times \mathbf{b}) = \rho \mathbf{a} - \kappa \mathbf{b}$. Conversely, if $\mathbf{a} \times \mathbf{b} \neq 0$ then $\mathbf{a}, \mathbf{b}$ are independent, so the identity forces $\mathbf{r} \cdot \mathbf{b} = \rho$ and $\mathbf{r} \cdot \mathbf{a} = \kappa$. If $\mathbf{a} \times \mathbf{b} = 0$, the planes are either identical (compatible constants) or parallel disjoint (incompatible constants).

Problem 3 – Matrices from ε_{ijk} and a unit vector

Exact statement. Show that

$$M_{ij} = \delta_{ij} + \varepsilon_{ijk} n_k, \quad N_{ij} = \delta_{ij} - \varepsilon_{ijk} n_k + n_i n_j$$

obey $N_{ij} M_{jk} = 2\delta_{ik}$ if $n_i n_i = 1$. Verify that

$$\mathbf{y} = \mathbf{x} + \mathbf{x} \times \mathbf{n} \iff y_i = M_{ij} x_j,$$

and use these results to find $\mathbf{x}$ in terms of $\mathbf{y}$, given that $\mathbf{n}$ is a unit vector.

Solution. Expand

$$(NM)_{ik} = N_{ij} M_{jk} = (\delta_{ij} - \varepsilon_{ij\ell} n_\ell + n_i n_j)(\delta_{jk} + \varepsilon_{jkm} n_m).$$

The mixed ε terms cancel, and $n_i n_j \varepsilon_{jkm} n_m = 0$. Using

$$\varepsilon_{ij\ell} \varepsilon_{jkm} = -\delta_{ik} \delta_{\ell m} + \delta_{im} \delta_{\ell k},$$

we get

$$\varepsilon_{ij\ell} \varepsilon_{jkm} n_\ell n_m = -\delta_{ik} (n_\ell n_\ell) + n_i n_k = -\delta_{ik} + n_i n_k$$

since $n_\ell n_\ell = 1$. Hence

$$(NM)_{ik} = \delta_{ik} - (-\delta_{ik} + n_i n_k) + n_i n_k = 2\delta_{ik}.$$

Also

$$y_i = x_i + \varepsilon_{ijk} x_j n_k = (\delta_{ij} + \varepsilon_{ijk} n_k) x_j = M_{ij} x_j.$$

Multiplying by N gives $N_{ij} y_j = 2x_i$, so

$$x_i = \frac{1}{2} N_{ij} y_j = \frac{1}{2} (y_i - \varepsilon_{ijk} y_j n_k + n_i (n_j y_j)),$$

i.e.

$$\mathbf{x} = \frac{1}{2} (\mathbf{y} - \mathbf{y} \times \mathbf{n} + \mathbf{n}(\mathbf{n} \cdot \mathbf{y})).$$

Problem 4 – Maximal independent subsets in $\mathbb{R}^4$

Exact statement. The set X consists of six vectors in $\mathbb{R}^4$:

$$(1, 1, 0, 0), (1, 0, 1, 0), (1, 0, 0, 1), (0, 1, 1, 0), (0, 1, 0, 1), (0, 0, 1, 1).$$

Find two different subsets $Y \subset X$ whose members are linearly independent, each of which yields a linearly dependent subset of X whenever any element $v \in X$ with $v \notin Y$ is adjoined to Y .

Solution. Write the vectors as $v_{ij} = e_i + e_j$ in the standard basis of $\mathbb{R}^4$:

$$v_{12}, v_{13}, v_{14}, v_{23}, v_{24}, v_{34}.$$

The span of X is all of $\mathbb{R}^4$ (e.g. $2e_1 = v_{13} + v_{14} - v_{34}$), so any 4 independent vectors from X form a basis of $\mathbb{R}^4$; adjoining any other vector from X then forces dependence.

Two valid choices are:

$$Y_1 = \{v_{12}, v_{13}, v_{14}, v_{23}\}, \quad Y_2 = \{v_{12}, v_{13}, v_{14}, v_{34}\}.$$

For Y_1 the remaining vectors are

$$v_{24} = v_{14} + v_{23} - v_{13}, \quad v_{34} = v_{13} + v_{24} - v_{12},$$

so adding v_{24} or v_{34} makes dependence. For Y_2 the remaining vectors are

$$v_{23} = v_{12} - v_{14} + v_{34}, \quad v_{24} = v_{12} - v_{13} + v_{34},$$

so adding v_{23} or v_{24} makes dependence.

Problem 5 – A linear recurrence subspace in $\mathbb{R}^n$

Exact statement. Let V be the set of all vectors $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ ($n \geq 4$) such that

$$x_i + x_{i+1} + x_{i+2} + x_{i+3} = 0 \quad \text{for } i = 1, 2, \dots, n-3.$$

Show that V is a subspace of $\mathbb{R}^n$, and find a basis for V .

Solution. The defining conditions are homogeneous linear equations in $(x_1, \dots, x_n)$, hence V is closed under addition and scalar multiplication; therefore V is a subspace.

From

$$x_i + x_{i+1} + x_{i+2} + x_{i+3} = 0$$

we obtain the recurrence

$$x_{i+3} = -(x_i + x_{i+1} + x_{i+2}) \quad (i = 1, \dots, n-3).$$

Thus once (x_1, x_2, x_3) are chosen, all later coordinates are determined uniquely, so $\dim V = 3$.

Define three vectors $v^{(1)}, v^{(2)}, v^{(3)} \in \mathbb{R}^n$ by

$$v^{(1)} = (1, -1, 1, -1, 1, -1, \dots),$$

$$v^{(2)} = (1, 0, -1, 0, 1, 0, -1, 0, \dots),$$

$$v^{(3)} = (0, 1, 0, -1, 0, 1, 0, -1, \dots),$$

truncated to length n . In each case the sum of any four consecutive components is 0, hence $v^{(1)}, v^{(2)}, v^{(3)} \in V$.

Their first three entries are

$$(1, -1, 1), \quad (1, 0, -1), \quad (0, 1, 0),$$

which are linearly independent in $\mathbb{R}^3$, so $v^{(1)}, v^{(2)}, v^{(3)}$ are linearly independent in $\mathbb{R}^n$.

Finally, for any $\mathbf{x} \in V$ choose α, β, γ so that the first three coordinates match:

$$x_1 = \alpha + \beta, \quad x_2 = -\alpha + \gamma, \quad x_3 = \alpha - \beta,$$

hence

$$\alpha = \frac{x_1 + x_3}{2}, \quad \beta = \frac{x_1 - x_3}{2}, \quad \gamma = x_2 + \frac{x_1 + x_3}{2}.$$

Then $\mathbf{x}$ and $\alpha v^{(1)} + \beta v^{(2)} + \gamma v^{(3)}$ satisfy the same recurrence and agree in the first three entries, so they are equal. Therefore

$$V = \text{span}\{v^{(1)}, v^{(2)}, v^{(3)}\},$$

and $\{v^{(1)}, v^{(2)}, v^{(3)}\}$ is a basis of V .

Problem 6 – Cauchy–Schwarz and two applications

Exact statement. State the Cauchy–Schwarz inequality for vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ and give a necessary and sufficient condition for equality to hold.

(a) By considering suitable vectors in $\mathbb{R}^3$, or otherwise, show that

$$x^2 + y^2 + z^2 \geq yz + zx + xy$$

for any real numbers x, y, z .

(b) By considering suitable vectors in $\mathbb{R}^4$, or otherwise, show that

$$3(x^2 + y^2 + z^2 + 4) - 2(yz + zx + xy) - 4(x + y + z) = 0$$

holds for unique real values of x, y, z , to be determined.

Solution. Cauchy–Schwarz: for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$,

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

Equality holds if and only if $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent (including the case that one is zero).

(a) Let $\mathbf{u} = (x, y, z)$ and $\mathbf{v} = (y, z, x)$ in $\mathbb{R}^3$. Then

$$\mathbf{u} \cdot \mathbf{v} = xy + yz + zx, \quad \|\mathbf{u}\|^2 = \|\mathbf{v}\|^2 = x^2 + y^2 + z^2.$$

By Cauchy–Schwarz,

$$|xy + yz + zx| \leq \|\mathbf{u}\| \|\mathbf{v}\| = x^2 + y^2 + z^2,$$

hence $xy + yz + zx \leq x^2 + y^2 + z^2$. Equality forces $(x, y, z) = \lambda(y, z, x)$, which over $\mathbb{R}$ implies $x = y = z$ (or all zero).

(b) Write $s = x + y + z$ and $Q = x^2 + y^2 + z^2$. Since

$$xy + yz + zx = \frac{s^2 - Q}{2},$$

the equation becomes

$$3(Q + 4) - 2 \cdot \frac{s^2 - Q}{2} - 4s = 0 \iff 4Q - s^2 - 4s + 12 = 0.$$

Let $m = s/3$ and set $p = x - m$, $q = y - m$, $r = z - m$ so that $p + q + r = 0$ and

$$Q = 3m^2 + (p^2 + q^2 + r^2).$$

Substitution gives

$$4Q - s^2 - 4s + 12 = 3(m - 2)^2 + 4(p^2 + q^2 + r^2).$$

Hence the equation holds iff

$$3(m - 2)^2 + 4(p^2 + q^2 + r^2) = 0,$$

so $m = 2$ and $p = q = r = 0$, i.e.

$$x = y = z = 2.$$

Therefore the unique real solution is $(x, y, z) = (2, 2, 2)$.

section*Problem 5 – A linear recurrence subspace in $\mathbb{R}^n$

Exact statement. Let V be the set of all vectors $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ ($n \geq 4$) such that

$$x_i + x_{i+1} + x_{i+2} + x_{i+3} = 0 \quad \text{for } i = 1, 2, \dots, n-3.$$

Show that V is a subspace of $\mathbb{R}^n$, and find a basis for V .

Solution. The defining conditions are homogeneous linear equations in $(x_1, \dots, x_n)$, hence V is closed under addition and scalar multiplication; therefore V is a subspace.

From

$$x_i + x_{i+1} + x_{i+2} + x_{i+3} = 0$$

we obtain the recurrence

$$x_{i+3} = -(x_i + x_{i+1} + x_{i+2}) \quad (i = 1, \dots, n-3).$$

Thus once (x_1, x_2, x_3) are chosen, all later coordinates are determined uniquely, so $\dim V = 3$.

Define three vectors $v^{(1)}, v^{(2)}, v^{(3)} \in \mathbb{R}^n$ by

$$v^{(1)} = (1, -1, 1, -1, 1, -1, \dots),$$

$$v^{(2)} = (1, 0, -1, 0, 1, 0, -1, 0, \dots),$$

$$v^{(3)} = (0, 1, 0, -1, 0, 1, 0, -1, \dots),$$

truncated to length n . In each case the sum of any four consecutive components is 0, hence $v^{(1)}, v^{(2)}, v^{(3)} \in V$.

Their first three entries are

$$(1, -1, 1), \quad (1, 0, -1), \quad (0, 1, 0),$$

which are linearly independent in $\mathbb{R}^3$, so $v^{(1)}, v^{(2)}, v^{(3)}$ are linearly independent in $\mathbb{R}^n$.

Finally, for any $\mathbf{x} \in V$ choose α, β, γ so that the first three coordinates match:

$$x_1 = \alpha + \beta, \quad x_2 = -\alpha + \gamma, \quad x_3 = \alpha - \beta,$$

hence

$$\alpha = \frac{x_1 + x_3}{2}, \quad \beta = \frac{x_1 - x_3}{2}, \quad \gamma = x_2 + \frac{x_1 + x_3}{2}.$$

Then $\mathbf{x}$ and $\alpha v^{(1)} + \beta v^{(2)} + \gamma v^{(3)}$ satisfy the same recurrence and agree in the first three entries, so they are equal. Therefore

$$V = \text{span}\{v^{(1)}, v^{(2)}, v^{(3)}\},$$

and $\{v^{(1)}, v^{(2)}, v^{(3)}\}$ is a basis of V .

section*Problem 6 – Cauchy–Schwarz and two applications

Exact statement. State the Cauchy–Schwarz inequality for vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ and give a necessary and sufficient condition for equality to hold.

(a) By considering suitable vectors in $\mathbb{R}^3$, or otherwise, show that

$$x^2 + y^2 + z^2 \geq yz + zx + xy$$

for any real numbers x, y, z .

(b) By considering suitable vectors in $\mathbb{R}^4$, or otherwise, show that

$$3(x^2 + y^2 + z^2 + 4) - 2(yz + zx + xy) - 4(x + y + z) = 0$$

holds for unique real values of x, y, z , to be determined.

Solution. Cauchy–Schwarz: for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$,

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

Equality holds if and only if $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent (including the case that one is zero).

(a) Let $\mathbf{u} = (x, y, z)$ and $\mathbf{v} = (y, z, x)$ in $\mathbb{R}^3$. Then

$$\mathbf{u} \cdot \mathbf{v} = xy + yz + zx, \quad \|\mathbf{u}\|^2 = \|\mathbf{v}\|^2 = x^2 + y^2 + z^2.$$

By Cauchy–Schwarz,

$$|xy + yz + zx| \leq \|\mathbf{u}\| \|\mathbf{v}\| = x^2 + y^2 + z^2,$$

hence $xy + yz + zx \leq x^2 + y^2 + z^2$. Equality forces $(x, y, z) = \lambda(y, z, x)$, which over $\mathbb{R}$ implies $x = y = z$ (or all zero).

(b) Write $s = x + y + z$ and $Q = x^2 + y^2 + z^2$. Since

$$xy + yz + zx = \frac{s^2 - Q}{2},$$

the equation becomes

$$3(Q + 4) - 2 \cdot \frac{s^2 - Q}{2} - 4s = 0 \iff 4Q - s^2 - 4s + 12 = 0.$$

Let $m = s/3$ and set $p = x - m$, $q = y - m$, $r = z - m$ so that $p + q + r = 0$ and

$$Q = 3m^2 + (p^2 + q^2 + r^2).$$

Substitution gives

$$4Q - s^2 - 4s + 12 = 3(m - 2)^2 + 4(p^2 + q^2 + r^2).$$

Hence the equation holds iff

$$3(m - 2)^2 + 4(p^2 + q^2 + r^2) = 0,$$

so $m = 2$ and $p = q = r = 0$, i.e.

$$x = y = z = 2.$$

Therefore the unique real solution is $(x, y, z) = (2, 2, 2)$.

Alternative basis (constructed from initial conditions). From

$$x_i + x_{i+1} + x_{i+2} + x_{i+3} = 0$$

we obtain the recurrence

$$x_{i+3} = -(x_i + x_{i+1} + x_{i+2}),$$

so every $\mathbf{x} \in V$ is uniquely determined by (x_1, x_2, x_3) .

Define $\mathbf{b}^{(1)}, \mathbf{b}^{(2)}, \mathbf{b}^{(3)} \in \mathbb{R}^n$ as the unique vectors satisfying the recurrence with initial triples

$$\mathbf{b}_{1..3}^{(1)} = (1, 0, 0), \quad \mathbf{b}_{1..3}^{(2)} = (0, 1, 0), \quad \mathbf{b}_{1..3}^{(3)} = (0, 0, 1).$$

Then (by iterating the recurrence) these are periodic of period 4:

$$\mathbf{b}^{(1)} = (1, 0, 0, -1, 1, 0, 0, -1, \dots),$$

$$\mathbf{b}^{(2)} = (0, 1, 0, -1, 0, 1, 0, -1, \dots),$$

$$\mathbf{b}^{(3)} = (0, 0, 1, -1, 0, 0, 1, -1, \dots),$$

truncated to length n .

They are linearly independent because their first three entries are the standard basis of $\mathbb{R}^3$. Moreover, for any $\mathbf{x} \in V$, the vector

$$x_1 \mathbf{b}^{(1)} + x_2 \mathbf{b}^{(2)} + x_3 \mathbf{b}^{(3)}$$

satisfies the same recurrence and has the same first three components as $\mathbf{x}$, hence equals $\mathbf{x}$. Therefore $\{\mathbf{b}^{(1)}, \mathbf{b}^{(2)}, \mathbf{b}^{(3)}\}$ is a basis of V and $\dim V = 3$.

Problem 7 – Projection and cross-product maps (unit vector $\mathbf{n} \in \mathbb{R}^3$)

Exact statement. Let $\mathbf{n}$ be a unit vector in $\mathbb{R}^3$. Identify the image and kernel (null space) of each of the following linear maps $\mathbb{R}^3 \rightarrow \mathbb{R}^3$:

$$(a) \quad T : \mathbf{x} \mapsto \mathbf{x}' = \mathbf{x} - (\mathbf{x} \cdot \mathbf{n})\mathbf{n}, \quad (b) \quad Q : \mathbf{x} \mapsto \mathbf{x}' = \mathbf{n} \times \mathbf{x}.$$

Show that $T^2 = T$ and interpret the map T geometrically. Interpret the maps Q^2 and $Q^3 + Q$, and show that $Q^4 = T$.

Solution. For T , write $\mathbf{x} = \mathbf{x}_\perp + (\mathbf{x} \cdot \mathbf{n})\mathbf{n}$ with $\mathbf{x}_\perp \cdot \mathbf{n} = 0$. Then

$$T(\mathbf{x}) = \mathbf{x} - (\mathbf{x} \cdot \mathbf{n})\mathbf{n} = \mathbf{x}_\perp.$$

Hence

$$\ker T = \text{span}\{\mathbf{n}\}, \quad \text{Im } T = \mathbf{n}^\perp = \{\mathbf{y} : \mathbf{y} \cdot \mathbf{n} = 0\}.$$

Also $T(\mathbf{x}) \cdot \mathbf{n} = 0$, so

$$T^2(\mathbf{x}) = T(T(\mathbf{x})) = T(\mathbf{x}),$$

i.e. $T^2 = T$. Geometrically, T is the orthogonal projection onto the plane $\mathbf{n}^\perp$.

For $Q(\mathbf{x}) = \mathbf{n} \times \mathbf{x}$, we have $\mathbf{n} \times \mathbf{x} = \mathbf{0}$ iff $\mathbf{x}$ is parallel to $\mathbf{n}$, so

$$\ker Q = \text{span}\{\mathbf{n}\}.$$

Also $\mathbf{n} \cdot (\mathbf{n} \times \mathbf{x}) = 0$, so $\text{Im } Q \subseteq \mathbf{n}^\perp$. Conversely, if $\mathbf{y} \in \mathbf{n}^\perp$, take $\mathbf{x} = -\mathbf{n} \times \mathbf{y}$ and use

$$\mathbf{n} \times (\mathbf{n} \times \mathbf{y}) = \mathbf{n}(\mathbf{n} \cdot \mathbf{y}) - \mathbf{y}(\mathbf{n} \cdot \mathbf{n}) = -\mathbf{y}$$

to get $Q(\mathbf{x}) = \mathbf{y}$. Thus $\text{Im } Q = \mathbf{n}^\perp$.

Moreover,

$$Q^2(\mathbf{x}) = \mathbf{n} \times (\mathbf{n} \times \mathbf{x}) = \mathbf{n}(\mathbf{n} \cdot \mathbf{x}) - \mathbf{x} = -(\mathbf{x} - (\mathbf{x} \cdot \mathbf{n})\mathbf{n}) = -T(\mathbf{x}),$$

so $Q^2 = -T$. Then

$$Q^3 = Q(Q^2) = Q(-T) = -QT.$$

But $Q(\mathbf{x}) = Q(T\mathbf{x})$ since Q kills the component parallel to $\mathbf{n}$, hence $QT = Q$ and $Q^3 = -Q$, i.e. $Q^3 + Q = 0$. Finally,

$$Q^4 = (Q^2)^2 = (-T)^2 = T^2 = T.$$

Problem 8 – Images and kernels of two maps on $\mathbb{R}^3$

Exact statement. Give a geometrical description of the images and kernels of each of the following linear maps on $\mathbb{R}^3$:

$$(a) \quad T : (x, y, z) \mapsto (x + 2y + z, x + 2y + z, 2x + 4y + 2z),$$

$$(b) \quad S : (x, y, z) \mapsto (x + 2y + 3z, x - y + z, x + 5y + 5z).$$

Solution. (a) Put $s = x + 2y + z$. Then

$$T(x, y, z) = (s, s, 2s) = s(1, 1, 2).$$

Hence $\text{Im } T = \text{span}\{(1, 1, 2)\}$ (a line through the origin) and

$$\ker T = \{(x, y, z) : x + 2y + z = 0\}$$

(a plane through the origin).

(b) The matrix of S is

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & -1 & 1 \\ 1 & 5 & 5 \end{pmatrix},$$

which has rank 2, so $\dim \ker S = 1$ and $\dim \text{Im } S = 2$. Solving $A(x, y, z)^T = 0$ gives a kernel direction $(-5, -2, 3)$, hence

$$\ker S = \text{span}\{(-5, -2, 3)\}.$$

A basis for the image is given by independent columns, e.g.

$$(1, 1, 1), \quad (2, -1, 5),$$

so $\text{Im } S = \text{span}\{(1, 1, 1), (2, -1, 5)\}$, a plane through the origin. A normal vector is $(1, 1, 1) \times (2, -1, 5) = (2, -1, -1)$, so equivalently

$$\text{Im } S = \{(X, Y, Z) : 2X - Y - Z = 0\}.$$

Problem 9 – Image and kernel in $\mathbb{R}^4$ depending on parameters a, b

Exact statement. A linear map $\mathbb{R}^4 \rightarrow \mathbb{R}^4$ is defined by $\mathbf{x} \mapsto M\mathbf{x}$, where

$$M = \begin{pmatrix} a & a & b & a \\ a & a & b & 0 \\ a & b & a & b \\ a & b & a & 0 \end{pmatrix}.$$

Find the image and kernel of this map for all real values of a and b .

Solution. *Case 1:* $a = b = 0$. Then $M = 0$, so $\text{Im } M = \{0\}$ and $\ker M = \mathbb{R}^4$.

Case 2: $a = b \neq 0$. Then $\text{rank}(M) = 2$ and the equations $M\mathbf{x} = 0$ reduce to

$$x_4 = 0, \quad x_1 + x_2 + x_3 = 0.$$

Hence

$$\ker M = \{(x_1, x_2, x_3, 0) : x_1 + x_2 + x_3 = 0\} = \text{span}\{(-1, 1, 0, 0), (-1, 0, 1, 0)\}.$$

Also the columns satisfy $c_1 = c_2 = c_3 = a(1, 1, 1, 1)^T$ and $c_4 = a(1, 0, 1, 0)^T$, so

$$\text{Im } M = \text{span}\{(1, 1, 1, 1), (1, 0, 1, 0)\} = \{(u, v, u, v) : u, v \in \mathbb{R}\}.$$

Case 3: $a \neq b$. Then $\text{rank}(M) = 3$ and $\dim \ker M = 1$. One generator of the kernel is

$$\mathbf{k} = (-(a+b), a, a, 0), \quad \text{so} \quad \ker M = \text{span}\{\mathbf{k}\}.$$

Since $\dim(\text{Im } M) = 3$, the image is a hyperplane in $\mathbb{R}^4$. A vector orthogonal to all columns is

$$\mathbf{w} = (b, -b, -a, a),$$

so

$$\text{Im } M = \{\mathbf{y} \in \mathbb{R}^4 : \mathbf{w} \cdot \mathbf{y} = 0\} = \{(y_1, y_2, y_3, y_4) : b(y_1 - y_2) + a(y_4 - y_3) = 0\}.$$

Problem 10 – Rodrigues' rotation formula and extracting $\theta, \mathbf{n}$ from R

Exact statement. The linear map $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$\mathbf{x} \mapsto \mathbf{x}' = \cos \theta \mathbf{x} + (\mathbf{x} \cdot \mathbf{n})(1 - \cos \theta) \mathbf{n} - \sin \theta (\mathbf{x} \times \mathbf{n})$$

is a rotation by angle θ in a positive sense about the unit vector $\mathbf{n}$. Check this in the case $\mathbf{n} = (0, 0, 1)$.

Show that the expression above can be written as $\mathbf{x}' = R\mathbf{x}$, where R is a matrix with entries R_{ij} found explicitly (in terms of $\theta, n_i, \delta_{ij}, \varepsilon_{ijk}$). Hence show that

$$R_{ii} = 2 \cos \theta + 1, \quad \varepsilon_{ijk} R_{jk} = -2n_i \sin \theta.$$

Determine θ and $\mathbf{n}$ for

$$R = \frac{1}{3} \begin{pmatrix} 2 & -1 & 2 \\ 2 & 2 & -1 \\ -1 & 2 & 2 \end{pmatrix}.$$

Solution. For $\mathbf{n} = (0, 0, 1)$ and $\mathbf{x} = (x, y, z)$, we have $\mathbf{x} \cdot \mathbf{n} = z$ and

$$\mathbf{x} \times \mathbf{n} = (y, -x, 0),$$

so

$$\mathbf{x}' = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta, z),$$

the standard rotation about the z -axis.

In components,

$$x'_i = \cos \theta x_i + (1 - \cos \theta)(n_j x_j) n_i - \sin \theta \varepsilon_{ijk} x_j n_k,$$

so $x'_i = R_{ij} x_j$ with

$$R_{ij} = \cos \theta \delta_{ij} + (1 - \cos \theta) n_i n_j - \sin \theta \varepsilon_{ijk} n_k.$$

Taking the trace gives

$$R_{ii} = \cos \theta \delta_{ii} + (1 - \cos \theta) n_i n_i - \sin \theta \varepsilon_{iik} n_k = 3 \cos \theta + (1 - \cos \theta) \cdot 1 = 1 + 2 \cos \theta.$$

Also,

$$\varepsilon_{ijk} R_{jk} = -\sin \theta \varepsilon_{ijk} \varepsilon_{jkm} n_m = -\sin \theta (2\delta_{im}) n_m = -2n_i \sin \theta.$$

For the given matrix, $\text{tr}(R) = \frac{1}{3}(2 + 2 + 2) = 2$, hence

$$1 + 2 \cos \theta = 2 \quad \Rightarrow \quad \cos \theta = \frac{1}{2} \quad \Rightarrow \quad \theta = \frac{\pi}{3}.$$

Next,

$$(\varepsilon_{ijk} R_{jk})_i = (R_{32} - R_{23}, R_{13} - R_{31}, R_{21} - R_{12}) = (1, 1, 1).$$

Since $\varepsilon_{ijk} R_{jk} = -2\mathbf{n} \sin \theta$ and $\sin(\pi/3) = \sqrt{3}/2$, we get

$$(1, 1, 1) = -(\sqrt{3}) \mathbf{n} \quad \Rightarrow \quad \mathbf{n} = -\frac{1}{\sqrt{3}}(1, 1, 1).$$

Problem 11 – 2×2 matrices for plane transformations and complex multiplication

Exact statement.

- (a) Give examples of 2×2 real matrices representing the following types of transformations in $\mathbb{R}^2$: (i) reflection; (ii) dilation (or scaling); (iii) shear; and (iv) rotation. Which of these types of transformation are always represented by a 2×2 matrix with determinant +1? For which types (i)–(iv) do transformations A and B of the same type obey $AB = BA$ in general?

- (b) A linear map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $\mathbf{x} \mapsto \mathbf{x}' = M\mathbf{x}$ is defined by $z' = cz$ where $z = x_1 + ix_2$, $z' = x'_1 + ix'_2$ and $c = a + ib$ is a fixed complex number. Find the real 2×2 matrix M in terms of a and b . Which types of transformations (i)–(iv) can be obtained for particular choices of $c = a + ib$?

Solution.

- (a) Examples:

$$\text{reflection across } x\text{-axis: } \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (\det = -1),$$

$$\text{dilation by } s: \begin{pmatrix} s & 0 \\ 0 & s \end{pmatrix} \quad (\det = s^2),$$

$$\text{shear (horizontal) by } k: \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \quad (\det = 1),$$

$$\text{rotation by } \theta: \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad (\det = 1).$$

Thus rotations and (standard) shears always have determinant $+1$; reflections always have determinant -1 ; dilations have determinant s^2 (not always 1).

Commutation: dilations commute (scalar matrices), and rotations in $\mathbb{R}^2$ commute: $Rot_\alpha Rot_\beta = Rot_{\alpha+\beta} = Rot_\beta Rot_\alpha$. Reflections do not commute in general, and shears do not commute in general (only shears along the same direction commute).

- (b) Compute

$$z' = (a + ib)(x_1 + ix_2) = (ax_1 - bx_2) + i(bx_1 + ax_2).$$

Hence

$$\begin{pmatrix} x'_1 \\ x'_2 \end{pmatrix} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad M = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \quad \det M = a^2 + b^2.$$

Writing $c = re^{i\varphi}$ gives $M = r \begin{pmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{pmatrix}$, i.e. a rotation by φ and a uniform scaling by r . Thus one obtains dilations (take $b = 0$, $a > 0$) and rotations (take $a^2 + b^2 = 1$), but not reflections (which require $\det = -1$) nor shears.

Problem 12 – Reflections $H(\mathbf{n})$ and rotations $R(\mathbf{n}, \theta)$ in $\mathbb{R}^3$

Exact statement. Let $R(\mathbf{n}, \theta)$ be the matrix corresponding to a rotation with angle θ and axis $\mathbf{n}$, as given in (*) of Question 10. Let $H(\mathbf{n})$ be the matrix corresponding to reflection in a plane through the origin with unit normal $\mathbf{n}$, as defined by

$$\mathbf{x} \mapsto \mathbf{x}' = H(\mathbf{n})\mathbf{x} = \mathbf{x} - 2(\mathbf{x} \cdot \mathbf{n})\mathbf{n}.$$

In the following, $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the standard orthonormal basis vectors in $\mathbb{R}^3$.

- (a) Find explicitly the matrices $R(\mathbf{i}, \pi/2)$ and $R(\mathbf{j}, \pi/2)$ and check that $R(\mathbf{i}, \pi/2)R(\mathbf{j}, \pi/2) \neq R(\mathbf{j}, \pi/2)R(\mathbf{i}, \pi/2)$.
- (b) Show by both algebraic and geometrical means that the map $\mathbf{x} \mapsto \mathbf{x}' = -H(\mathbf{n})\mathbf{x}$ is a rotation through angle π about $\mathbf{n}$.
- (c) Given that $\mathbf{n}_{\pm} = \cos(\frac{\theta}{2})\mathbf{i} \pm \sin(\frac{\theta}{2})\mathbf{j}$, prove that

$$H(\mathbf{i})H(\mathbf{n}_{-}) = H(\mathbf{n}_{+})H(\mathbf{i}) = R(\mathbf{k}, \theta),$$

and draw diagrams to explain the geometrical meaning of this result.

Solution.

- (a) The 90° rotation about the x -axis is

$$R(\mathbf{i}, \frac{\pi}{2}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix},$$

and the 90° rotation about the y -axis is

$$R(\mathbf{j}, \frac{\pi}{2}) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

Then

$$R(\mathbf{i}, \frac{\pi}{2})R(\mathbf{j}, \frac{\pi}{2}) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{pmatrix} = R(\mathbf{j}, \frac{\pi}{2})R(\mathbf{i}, \frac{\pi}{2}),$$

so they do not commute.

- (b) Decompose $\mathbf{x} = \mathbf{x}_{\parallel} + \mathbf{x}_{\perp}$ where $\mathbf{x}_{\parallel} = (\mathbf{x} \cdot \mathbf{n})\mathbf{n}$ and $\mathbf{x}_{\perp} \cdot \mathbf{n} = 0$. Then

$$H(\mathbf{n})\mathbf{x} = \mathbf{x} - 2\mathbf{x}_{\parallel} = \mathbf{x}_{\perp} - \mathbf{x}_{\parallel},$$

so

$$(-H(\mathbf{n}))\mathbf{x} = -\mathbf{x}_{\perp} + \mathbf{x}_{\parallel}.$$

Thus $-H(\mathbf{n})$ fixes the axis direction $\mathbf{n}$ and reverses every vector orthogonal to $\mathbf{n}$, i.e. it is a rotation by angle π about axis $\mathbf{n}$.

- (c) Let $\alpha = \theta/2$, so

$$\mathbf{n}_{-} = (\cos \alpha, -\sin \alpha, 0), \quad \mathbf{n}_{+} = (\cos \alpha, \sin \alpha, 0).$$

Using $H(\mathbf{u}) = I - 2\mathbf{u}\mathbf{u}^T$,

$$H(\mathbf{i}) = \text{diag}(-1, 1, 1),$$

and

$$H(\mathbf{n}_-) = \begin{pmatrix} 1 - 2\cos^2\alpha & 2\cos\alpha\sin\alpha & 0 \\ 2\cos\alpha\sin\alpha & 1 - 2\sin^2\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -\cos(2\alpha) & \sin(2\alpha) & 0 \\ \sin(2\alpha) & \cos(2\alpha) & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Therefore

$$H(\mathbf{i})H(\mathbf{n}_-) = \begin{pmatrix} \cos(2\alpha) & -\sin(2\alpha) & 0 \\ \sin(2\alpha) & \cos(2\alpha) & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = R(\mathbf{k}, \theta).$$

Similarly one checks $H(\mathbf{n}_+)H(\mathbf{i}) = R(\mathbf{k}, \theta)$. Geometrically, $H(\mathbf{i})$ and $H(\mathbf{n}_-)$ are reflections in two planes through the origin intersecting in the $\mathbf{k}$ -axis; the angle between the planes is $\alpha = \theta/2$, hence their composition is a rotation about $\mathbf{k}$ by angle $2\alpha = \theta$.

Theory for Problems 4–12 (Linear algebra + geometry toolbox)

Subspaces, kernel/image, rank-nullity. A subset $V \subseteq \mathbb{R}^n$ is a subspace iff it contains 0 and is closed under addition and scalar multiplication. Equivalently, $V = \ker(A)$ for some matrix A , or $V = \text{span}\{v_1, \dots, v_k\}$.

For a linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with matrix A :

$$\ker L = \{x : Ax = 0\}, \quad \text{Im } L = \{Ax : x \in \mathbb{R}^n\} = \text{column space of } A.$$

Rank-nullity:

$$\dim(\ker L) + \dim(\text{Im } L) = n.$$

In $\mathbb{R}^3$ this often forces $\ker$ and Im to be lines/planes unless L is 0 or invertible.

Problem 4: combinatorial picture for the six vectors in $\mathbb{R}^4$

The six vectors are $e_i + e_j$ for $1 \leq i < j \leq 4$, which can be viewed as the edges of the complete graph K_4 on vertices $\{1, 2, 3, 4\}$. A useful rule of thumb:

- A set of such vectors is linearly independent exactly when the corresponding edges contain no cycle (a forest).
- A maximal independent set of size 3 corresponds to a spanning tree on 4 vertices.

Typical dependencies come from cycles, e.g.

$$(e_1 + e_2) + (e_3 + e_4) = (e_1 + e_3) + (e_2 + e_4).$$

Problem 5: subspace defined by a recurrence

The constraints

$$x_i + x_{i+1} + x_{i+2} + x_{i+3} = 0$$

form a homogeneous linear system, hence the solution set is a subspace. They yield the recurrence

$$x_{i+3} = -(x_i + x_{i+1} + x_{i+2}),$$

so the whole vector is determined by (x_1, x_2, x_3) and therefore $\dim V = 3$. The characteristic polynomial is

$$r^3 + r^2 + r + 1 = (r + 1)(r^2 + 1),$$

which explains the frequent appearance of 4-periodic patterns.

Problem 6: Cauchy–Schwarz as a positivity principle

Cauchy–Schwarz:

$$|\langle u, v \rangle| \leq \|u\| \|v\|,$$

with equality iff u and v are linearly dependent. A standard proof uses

$$\|u - tv\|^2 \geq 0 \quad \forall t \in \mathbb{R}$$

and discriminant ≤ 0 . Many inequalities reduce to “sum of squares”, e.g.

$$x^2 + y^2 + z^2 - (xy + yz + zx) = \frac{1}{2}((x - y)^2 + (y - z)^2 + (z - x)^2) \geq 0.$$

Problems 7–8: projections, cross-product operator, and geometry of rank

For unit n ,

$$T(x) = x - (x \cdot n)n$$

is orthogonal projection onto $n^\perp$:

$$\ker T = \text{span}\{n\}, \quad \text{Im } T = n^\perp, \quad T^2 = T.$$

The cross-product map

$$Q(x) = n \times x$$

is represented by the skew-symmetric matrix

$$[n]_\times = \begin{pmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{pmatrix}, \quad Q(x) = [n]_\times x.$$

Using $n \times (n \times x) = n(n \cdot x) - x$, one gets

$$Q^2 = -T, \quad Q^3 = -Q, \quad Q^4 = T.$$

For any $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $\text{Im } A$ is the column space and $\ker A$ solves $Ax = 0$; the rank determines whether the image is a line (rank 1), plane (rank 2), or all $\mathbb{R}^3$ (rank 3).

Problem 9: parameter-dependent rank

For $M(a, b) : \mathbb{R}^4 \rightarrow \mathbb{R}^4$,

$$\dim(\operatorname{Im} M) = \operatorname{rank}(M), \quad \dim(\ker M) = 4 - \operatorname{rank}(M).$$

To analyze all (a, b) one typically uses rank-preserving row/column operations and studies when key minors vanish; rank drops only on special parameter loci (e.g. lines such as $a = b$, or the origin).

Problem 10: rotations and axis-angle extraction

A rotation matrix satisfies $R^T R = I$ and $\det R = 1$. It has an axis n with $Rn = n$ and can be written (Rodrigues)

$$R = \cos \theta I + (1 - \cos \theta)nn^T - \sin \theta [n]_{\times}.$$

Then

$$\operatorname{tr}(R) = 1 + 2 \cos \theta,$$

and the antisymmetric part determines n via

$$R - R^T = -2 \sin \theta [n]_{\times}, \quad \text{equivalently} \quad (R_{32} - R_{23}, R_{13} - R_{31}, R_{21} - R_{12}) = -2 \sin \theta n.$$

Problem 11: plane transformations and complex multiplication

Orthogonal maps satisfy $A^T A = I$:

$$\det A = +1 \Rightarrow \text{rotation}, \quad \det A = -1 \Rightarrow \text{reflection (or reflection+rotation)}.$$

Uniform scaling is sI ; shears are typically unipotent (determinant 1) but not angle-preserving. Complex multiplication $z \mapsto cz$ with $c = a + ib$ corresponds to

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} = r \begin{pmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{pmatrix},$$

so it yields rotation + uniform scaling only (no shear, no reflection). In $\mathbb{R}^2$ rotations commute; scalings commute with everything; reflections and general shears do not commute in general.

Problem 12: Householder reflections and “two reflections = rotation”

Reflection in the plane through the origin with unit normal n is

$$H(n) = I - 2nn^T, \quad H(n)^2 = I, \quad \det H(n) = -1.$$

The composition of reflections in two planes through the origin is a rotation about their line of intersection; the rotation angle equals twice the angle between the planes (equivalently twice the angle between their normals). Moreover, $-H(n)$ fixes n and flips every vector orthogonal to n , hence it is a rotation by π about axis n .

Problem 1 — Distance bounds from a disk (triangle inequalities)

Exact statement. Let D be the interior of the circle $|z - 1 - i| = 1$. Show, by using suitable inequalities for $|z_1 \pm z_2|$, that if $z \in D$ then

$$\sqrt{5} - 1 < |z - 3| < \sqrt{5} + 1.$$

Obtain the same result geometrically [start by considering the line through the centre of the circle and the point 3].

Solution. Let $c = 1 + i$. Then $|z - c| < 1$ for $z \in D$. Write

$$z - 3 = (z - c) + (c - 3).$$

By the triangle inequality and reverse triangle inequality,

$$|z - 3| \leq |z - c| + |c - 3|, \quad |z - 3| \geq ||c - 3| - |z - c||.$$

Now

$$|c - 3| = |(1 + i) - 3| = |-2 + i| = \sqrt{4 + 1} = \sqrt{5}.$$

Hence, since $|z - c| < 1$,

$$|z - 3| < 1 + \sqrt{5}, \quad |z - 3| > \sqrt{5} - 1,$$

which gives $\sqrt{5} - 1 < |z - 3| < \sqrt{5} + 1$.

Geometrically: D is the open disk of radius 1 centered at c . The distance from the fixed point 3 to points in that disk ranges strictly between $|3 - c| - 1 = \sqrt{5} - 1$ and $|3 - c| + 1 = \sqrt{5} + 1$.

Problem 2 — Modulus–argument forms on the unit circle; a Cayley transform locus

Exact statement. Given $|z| = 1$ and $\arg z = \theta$, find both algebraically and geometrically the modulus–argument forms of (i) $1 + z$, (ii) $1 - \bar{z}$.

Show that the locus of w as z varies with $|z| = 1$, where w is given by

$$w^2 = \frac{1 - z}{1 + z},$$

is a pair of straight lines.

Solution. Since $|z| = 1$ and $\arg z = \theta$, write $z = e^{i\theta}$ and $\bar{z} = e^{-i\theta}$.

(i) **Modulus–argument of $1 + z$.**

$$1 + z = 1 + e^{i\theta} = e^{i\theta/2}(e^{-i\theta/2} + e^{i\theta/2}) = 2 \cos\left(\frac{\theta}{2}\right) e^{i\theta/2}.$$

So $|1 + z| = 2|\cos(\theta/2)|$, and an argument is $\theta/2$ (adjust by π if $\cos(\theta/2) < 0$).

Geometrically: $1 + z$ is the diagonal of the parallelogram with sides 1 and z ; its direction bisects the angle between them, hence angle $\theta/2$.

(ii) **Modulus–argument of $1 - \bar{z}$.**

$$1 - \bar{z} = 1 - e^{-i\theta} = e^{-i\theta/2}(e^{i\theta/2} - e^{-i\theta/2}) = 2i \sin\left(\frac{\theta}{2}\right) e^{-i\theta/2}.$$

Equivalently,

$$1 - \bar{z} = 2 \sin\left(\frac{\theta}{2}\right) e^{i(\pi/2 - \theta/2)}.$$

So $|1 - \bar{z}| = 2|\sin(\theta/2)|$, and an argument is $\pi/2 - \theta/2$ (adjust by π if $\sin(\theta/2) < 0$).

Geometrically: $|1 - \bar{z}| = |1 - e^{-i\theta}|$ is the chord length between 1 and $e^{-i\theta}$ on the unit circle.

Locus of w . For $z = e^{i\theta}$ (excluding $z = -1$),

$$\frac{1 - z}{1 + z} = \frac{1 - e^{i\theta}}{1 + e^{i\theta}} = \frac{e^{-i\theta/2} - e^{i\theta/2}}{e^{-i\theta/2} + e^{i\theta/2}} = \frac{-2i \sin(\theta/2)}{2 \cos(\theta/2)} = -i \tan\left(\frac{\theta}{2}\right).$$

Thus $w^2 \in i\mathbb{R}$ (purely imaginary). Write $w = u + iv$. Then

$$w^2 = (u^2 - v^2) + 2uvi$$

is purely imaginary iff $u^2 - v^2 = 0$, i.e. $v = \pm u$. Therefore w lies on the two straight lines

$$v = u \quad \text{or} \quad v = -u.$$

Problem 3 — Medians of a complex triangle are concurrent

Exact statement. Consider a triangle in the complex plane with vertices at 0, z_1 and z_2 . Write down an expression for the general point on the median through z_1 , and a similar expression for the general point on the median through z_2 . Show that the three medians of the triangle are concurrent.

Solution. The midpoint of the side from 0 to z_2 is $z_2/2$. Hence the median through z_1 is the line segment joining z_1 to $z_2/2$, so a general point on it is

$$m_1(t) = (1 - t)z_1 + t \frac{z_2}{2}, \quad t \in \mathbb{R}.$$

Similarly, the midpoint of the side from 0 to z_1 is $z_1/2$, so the median through z_2 is

$$m_2(s) = (1 - s)z_2 + s \frac{z_1}{2}, \quad s \in \mathbb{R}.$$

Let

$$g = \frac{z_1 + z_2}{3}.$$

Then

$$g = \frac{1}{3}z_1 + \frac{1}{3}z_2 = \left(1 - \frac{2}{3}\right)z_1 + \frac{2}{3} \cdot \frac{z_2}{2} = m_1\left(\frac{2}{3}\right),$$

so g lies on the median through z_1 . Similarly,

$$g = \left(1 - \frac{2}{3}\right)z_2 + \frac{2}{3} \cdot \frac{z_1}{2} = m_2\left(\frac{2}{3}\right),$$

so g lies on the median through z_2 . The median through 0 goes to the midpoint $(z_1 + z_2)/2$, and

$$g = \frac{2}{3} \cdot \frac{z_1 + z_2}{2},$$

so g lies on that median too. Hence all three medians meet at g , so they are concurrent.

Problem 4 — Fifth roots of unity factorization and a trig identity

Exact statement. Express

$$I = \frac{z^5 - 1}{z - 1}$$

as a polynomial in z . By considering the complex fifth root of unity ω , obtain the four factors of I linear in z . Hence write I as the product of two real quadratic factors. By considering the term in z^2 in the identity so obtained for I , show that

$$4 \cos \frac{\pi}{5} \sin \frac{\pi}{10} = 1.$$

Solution. For $z \neq 1$,

$$\frac{z^5 - 1}{z - 1} = z^4 + z^3 + z^2 + z + 1,$$

so as a polynomial,

$$I = z^4 + z^3 + z^2 + z + 1.$$

Let $\omega = e^{2\pi i/5}$. Then

$$z^5 - 1 = (z - 1)(z - \omega)(z - \omega^2)(z - \omega^3)(z - \omega^4),$$

hence

$$I = (z - \omega)(z - \omega^2)(z - \omega^3)(z - \omega^4).$$

Pair conjugates $\omega^4 = \bar{\omega}$ and $\omega^3 = \bar{\omega}^2$:

$$(z - \omega)(z - \omega^4) = z^2 - (\omega + \omega^4)z + 1 = z^2 - 2 \cos\left(\frac{2\pi}{5}\right)z + 1,$$

$$(z - \omega^2)(z - \omega^3) = z^2 - (\omega^2 + \omega^3)z + 1 = z^2 - 2 \cos\left(\frac{4\pi}{5}\right)z + 1.$$

Therefore

$$I = \left(z^2 - 2 \cos\left(\frac{2\pi}{5}\right)z + 1\right) \left(z^2 - 2 \cos\left(\frac{4\pi}{5}\right)z + 1\right).$$

Expanding $(z^2 - az + 1)(z^2 - bz + 1)$ gives the z^2 -coefficient $ab + 2$. Comparing with $I = z^4 + z^3 + z^2 + z + 1$ yields

$$ab + 2 = 1 \implies ab = -1,$$

with $a = 2 \cos(2\pi/5)$, $b = 2 \cos(4\pi/5)$. Hence

$$4 \cos\left(\frac{2\pi}{5}\right) \cos\left(\frac{4\pi}{5}\right) = -1.$$

Using $\cos(4\pi/5) = -\cos(\pi/5)$ and $\cos(2\pi/5) = \sin(\pi/10)$, we get

$$4 \cos\left(\frac{\pi}{5}\right) \sin\left(\frac{\pi}{10}\right) = 1.$$

Problem 5 — Solve $\sin z = 2$ in $\mathbb{C}$

Exact statement. Find all complex numbers z that satisfy $\sin z = 2$.

Solution. Write $z = x + iy$. Then

$$\sin(x + iy) = \sin x \cosh y + i \cos x \sinh y.$$

We need $\sin z = 2$ real, so $\cos x \sinh y = 0$. Since $\sin z \neq 0$, we must have $\cos x = 0$, hence

$$x = \frac{\pi}{2} + k\pi.$$

Then $\sin x = (-1)^k$, so

$$\sin z = (-1)^k \cosh y.$$

To equal $2 > 0$, we need k even, i.e. $x = \frac{\pi}{2} + 2\pi n$. Then $\cosh y = 2$, so

$$y = \pm \operatorname{arcosh}(2) = \pm \ln(2 + \sqrt{3}).$$

Therefore all solutions are

$$z = \frac{\pi}{2} + 2\pi n \pm i \ln(2 + \sqrt{3}), \quad n \in \mathbb{Z}.$$

Problem 6 — Apollonius circle; a coaxal family and orthogonal intersections

Exact statement. (a) Let $z, a, b \in \mathbb{C}$ ($a \neq b$) correspond to points P, A, B in the Argand diagram. Let C_λ be the locus of P defined by

$$\frac{|PA|}{|PB|} = \lambda,$$

where λ is a fixed positive real constant. Show that C_λ is a circle if $\lambda \neq 1$, and find its centre and radius. What happens if $\lambda = 1$?

(b) For the case $a = -b = p$ with $p \in \mathbb{R}$, and for each fixed $\mu \in \mathbb{R}$, show that

$$S_\mu = \left\{ z \in \mathbb{C} : |z - i\mu| = \sqrt{p^2 + \mu^2} \right\}$$

is a circle passing through A and B with its centre on the perpendicular bisector of AB . Show that the circles C_λ and S_μ intersect orthogonally for all λ, μ .

Solution. (a) The condition $|PA|/|PB| = \lambda$ is

$$|z - a| = \lambda|z - b| \iff |z - a|^2 = \lambda^2|z - b|^2.$$

Expand:

$$|z|^2 - z\bar{a} - \bar{z}a + |a|^2 = \lambda^2(|z|^2 - z\bar{b} - \bar{z}b + |b|^2).$$

So

$$(1 - \lambda^2)|z|^2 - z(\bar{a} - \lambda^2\bar{b}) - \bar{z}(a - \lambda^2b) + (|a|^2 - \lambda^2|b|^2) = 0.$$

If $\lambda \neq 1$, divide by $1 - \lambda^2$ and complete the square:

$$\left| z - \frac{a - \lambda^2b}{1 - \lambda^2} \right|^2 = \frac{\lambda^2|a - b|^2}{(1 - \lambda^2)^2}.$$

Hence C_λ is a circle with centre

$$c_\lambda = \frac{a - \lambda^2 b}{1 - \lambda^2}$$

and radius

$$r_\lambda = \frac{\lambda|a - b|}{|1 - \lambda^2|}.$$

If $\lambda = 1$, the condition is $|z - a| = |z - b|$: the locus is the perpendicular bisector of AB .

(b) Here $a = p$, $b = -p$ with $p \in \mathbb{R}$. The set

$$S_\mu : |z - i\mu| = \sqrt{p^2 + \mu^2}$$

is a circle with centre $i\mu$, which lies on the imaginary axis (the perpendicular bisector of AB). It passes through $A = p$ and $B = -p$ since

$$|p - i\mu| = \sqrt{p^2 + \mu^2}, \quad |-p - i\mu| = \sqrt{p^2 + \mu^2}.$$

From (a),

$$c_\lambda = \frac{p - \lambda^2(-p)}{1 - \lambda^2} = p \frac{1 + \lambda^2}{1 - \lambda^2} \in \mathbb{R}, \quad r_\lambda = \frac{\lambda|2p|}{|1 - \lambda^2|}.$$

Two circles meet orthogonally iff the squared distance between centres equals the sum of squared radii:

$$|c_\lambda - i\mu|^2 = r_\lambda^2 + (p^2 + \mu^2).$$

But

$$|c_\lambda - i\mu|^2 = c_\lambda^2 + \mu^2,$$

so it suffices to show $c_\lambda^2 = r_\lambda^2 + p^2$. Indeed,

$$c_\lambda^2 = p^2 \frac{(1 + \lambda^2)^2}{(1 - \lambda^2)^2}, \quad r_\lambda^2 = \frac{4\lambda^2 p^2}{(1 - \lambda^2)^2},$$

hence

$$c_\lambda^2 - r_\lambda^2 = \frac{p^2((1 + \lambda^2)^2 - 4\lambda^2)}{(1 - \lambda^2)^2} = \frac{p^2(1 - \lambda^2)^2}{(1 - \lambda^2)^2} = p^2.$$

Therefore C_λ and S_μ intersect orthogonally for all $\lambda \neq 1$. If $\lambda = 1$, then C_1 is the imaginary axis, which passes through the centre $i\mu$ of S_μ , so it meets S_μ orthogonally as well.

Problem 7 — Concurrency of the altitudes (orthocenter) via vectors

Exact statement. Show by vector methods that the altitudes of a triangle are concurrent. [Hint: let the altitudes AD, BE of $\triangle ABC$ meet at H , and show that CH is perpendicular to AB].

Solution. Let the position vectors of A, B, C, H be $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{h}$. Since H lies on the altitude from A , we have $AH \perp BC$, i.e.

$$(\mathbf{h} - \mathbf{a}) \cdot (\mathbf{c} - \mathbf{b}) = 0 \implies \mathbf{h} \cdot (\mathbf{c} - \mathbf{b}) = \mathbf{a} \cdot (\mathbf{c} - \mathbf{b}).$$

Since H lies on the altitude from B , we have $BH \perp AC$, i.e.

$$(\mathbf{h} - \mathbf{b}) \cdot (\mathbf{c} - \mathbf{a}) = 0 \implies \mathbf{h} \cdot (\mathbf{c} - \mathbf{a}) = \mathbf{b} \cdot (\mathbf{c} - \mathbf{a}).$$

Now write $\mathbf{b} - \mathbf{a} = (\mathbf{b} - \mathbf{c}) + (\mathbf{c} - \mathbf{a})$. Then

$$\mathbf{h} \cdot (\mathbf{b} - \mathbf{a}) = \mathbf{h} \cdot (\mathbf{b} - \mathbf{c}) + \mathbf{h} \cdot (\mathbf{c} - \mathbf{a}) = -\mathbf{h} \cdot (\mathbf{c} - \mathbf{b}) + \mathbf{h} \cdot (\mathbf{c} - \mathbf{a}).$$

Substitute the relations above:

$$\mathbf{h} \cdot (\mathbf{b} - \mathbf{a}) = -\mathbf{a} \cdot (\mathbf{c} - \mathbf{b}) + \mathbf{b} \cdot (\mathbf{c} - \mathbf{a}) = \mathbf{c} \cdot (\mathbf{b} - \mathbf{a}).$$

Therefore

$$(\mathbf{h} - \mathbf{c}) \cdot (\mathbf{b} - \mathbf{a}) = 0,$$

so $CH \perp AB$. Hence H also lies on the altitude from C , and all three altitudes are concurrent.

Problem 8 — A vector equation with dot-product constraints and equality cases

Exact statement. In three dimensions, the vectors $\mathbf{x}, \mathbf{y}, \mathbf{a}$ satisfy

$$\mathbf{x} + \mathbf{y}(\mathbf{x} \cdot \mathbf{y}) = \mathbf{a}.$$

Show that

$$(\mathbf{x} \cdot \mathbf{y})^2 = \frac{|\mathbf{a}|^2 - |\mathbf{x}|^2}{2 + |\mathbf{y}|^2}.$$

Use an inequality involving $\mathbf{x} \cdot \mathbf{y}$ and the lengths of $\mathbf{x}$ and $\mathbf{y}$ to deduce that

$$|\mathbf{x}|(1 + |\mathbf{y}|^2) \geq |\mathbf{a}| \geq |\mathbf{x}|.$$

Explain the circumstances in which either of the inequalities above become equalities, and describe the relation between $\mathbf{x}, \mathbf{y}, \mathbf{a}$ in these circumstances.

Solution. Let $t = \mathbf{x} \cdot \mathbf{y}$. Then $\mathbf{a} = \mathbf{x} + t\mathbf{y}$. Taking norms,

$$|\mathbf{a}|^2 = |\mathbf{x} + t\mathbf{y}|^2 = |\mathbf{x}|^2 + 2t(\mathbf{x} \cdot \mathbf{y}) + t^2|\mathbf{y}|^2 = |\mathbf{x}|^2 + t^2(2 + |\mathbf{y}|^2).$$

Hence

$$t^2 = \frac{|\mathbf{a}|^2 - |\mathbf{x}|^2}{2 + |\mathbf{y}|^2}.$$

From this, $|\mathbf{a}|^2 \geq |\mathbf{x}|^2$, so $|\mathbf{a}| \geq |\mathbf{x}|$. Also by Cauchy–Schwarz, $|t| = |\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}||\mathbf{y}|$, so

$$\frac{|\mathbf{a}|^2 - |\mathbf{x}|^2}{2 + |\mathbf{y}|^2} = t^2 \leq |\mathbf{x}|^2|\mathbf{y}|^2.$$

Therefore

$$|\mathbf{a}|^2 \leq |\mathbf{x}|^2(1 + (2 + |\mathbf{y}|^2)|\mathbf{y}|^2) = |\mathbf{x}|^2(1 + |\mathbf{y}|^2)^2,$$

hence $|\mathbf{a}| \leq |\mathbf{x}|(1 + |\mathbf{y}|^2)$.

Equality $|\mathbf{a}| = |\mathbf{x}|$ holds iff $t = 0$, i.e. $\mathbf{x} \cdot \mathbf{y} = 0$, in which case $\mathbf{a} = \mathbf{x}$. Equality $|\mathbf{a}| = |\mathbf{x}|(1 + |\mathbf{y}|^2)$ holds iff $|\mathbf{x} \cdot \mathbf{y}| = |\mathbf{x}||\mathbf{y}|$, i.e. $\mathbf{x} \parallel \mathbf{y}$, and then $\mathbf{a} = \mathbf{x} + t\mathbf{y}$ is also parallel to them.

Problem 9 — Cross products in a triangle; sine rule; a converse

Exact statement. (a) In $\triangle ABC$, let $\overrightarrow{AB} = \mathbf{u}$, $\overrightarrow{BC} = \mathbf{v}$ and $\overrightarrow{CA} = \mathbf{w}$. Show that

$$\mathbf{u} \times \mathbf{v} = \mathbf{v} \times \mathbf{w} = \mathbf{w} \times \mathbf{u},$$

and hence obtain the sine rule for $\triangle ABC$.

(b) Given any three vectors $\mathbf{p}, \mathbf{q}, \mathbf{r}$ such that

$$\mathbf{p} \times \mathbf{q} = \mathbf{q} \times \mathbf{r} = \mathbf{r} \times \mathbf{p},$$

and $|\mathbf{p} \times \mathbf{q}| \neq 0$, show that

$$\mathbf{p} + \mathbf{q} + \mathbf{r} = \mathbf{0}.$$

Solution. (a) In a triangle,

$$\mathbf{u} + \mathbf{v} + \mathbf{w} = \mathbf{0} \implies \mathbf{w} = -(\mathbf{u} + \mathbf{v}).$$

Then

$$\mathbf{v} \times \mathbf{w} = \mathbf{v} \times (-(\mathbf{u} + \mathbf{v})) = -(\mathbf{v} \times \mathbf{u}) = \mathbf{u} \times \mathbf{v},$$

and similarly $\mathbf{w} \times \mathbf{u} = \mathbf{u} \times \mathbf{v}$. Hence $\mathbf{u} \times \mathbf{v} = \mathbf{v} \times \mathbf{w} = \mathbf{w} \times \mathbf{u}$.

Taking magnitudes and using $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin(\angle(\mathbf{a}, \mathbf{b}))$, we obtain

$$|\mathbf{u}||\mathbf{v}|\sin B = |\mathbf{v}||\mathbf{w}|\sin C = |\mathbf{w}||\mathbf{u}|\sin A,$$

which gives the sine rule

$$\frac{|BC|}{\sin A} = \frac{|CA|}{\sin B} = \frac{|AB|}{\sin C}.$$

(b) From $\mathbf{p} \times \mathbf{q} = \mathbf{q} \times \mathbf{r}$,

$$\mathbf{p} \times \mathbf{q} - \mathbf{q} \times \mathbf{r} = \mathbf{0} \implies \mathbf{p} \times \mathbf{q} + \mathbf{r} \times \mathbf{q} = \mathbf{0} \implies (\mathbf{p} + \mathbf{r}) \times \mathbf{q} = \mathbf{0},$$

so $\mathbf{p} + \mathbf{r} \parallel \mathbf{q}$. Also $\mathbf{r} \times \mathbf{p} = \mathbf{p} \times \mathbf{q}$ gives $(\mathbf{q} + \mathbf{r}) \times \mathbf{p} = \mathbf{0}$, so $\mathbf{q} + \mathbf{r} \parallel \mathbf{p}$. Since $|\mathbf{p} \times \mathbf{q}| \neq 0$, $\mathbf{p}$ and $\mathbf{q}$ are linearly independent, forcing $\mathbf{r} = -\mathbf{p} - \mathbf{q}$. Hence $\mathbf{p} + \mathbf{q} + \mathbf{r} = \mathbf{0}$.

Problem 10 — Geometric meaning of affine combinations (line and plane)

Exact statement. Interpret geometrically the equations

$$\mathbf{r} = (1 - \lambda)\mathbf{a} + \lambda\mathbf{b}, \quad \mathbf{r} = (1 - \lambda - \mu)\mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c},$$

where $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are the position vectors of points in three dimensions and λ, μ are scalar parameters that take any real values. For each equation, obtain an equivalent form that does not involve parameters λ or μ . How do your equations behave when the points $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are interchanged?

Solution.

$$\mathbf{r} = (1 - \lambda)\mathbf{a} + \lambda\mathbf{b} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a})$$

describes the line through the points with position vectors $\mathbf{a}$ and $\mathbf{b}$. A parameter-free form is

$$(\mathbf{r} - \mathbf{a}) \times (\mathbf{b} - \mathbf{a}) = \mathbf{0}.$$

Similarly,

$$\mathbf{r} = (1 - \lambda - \mu)\mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a}) + \mu(\mathbf{c} - \mathbf{a})$$

describes the plane through the points $\mathbf{a}, \mathbf{b}, \mathbf{c}$. A parameter-free form is

$$(\mathbf{r} - \mathbf{a}) \cdot ((\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a})) = 0.$$

Interchanging $\mathbf{a}, \mathbf{b}$ (or permuting $\mathbf{a}, \mathbf{b}, \mathbf{c}$) changes λ, μ but not the underlying line/plane; the parameter-free equations remain the same (up to an overall sign in the normal vector, which does not affect “= 0”).

Problem 11 — Lagrange’s identity, Jacobi identity, and scalar triple products

Exact statement. (a) Using the identity $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$, show that

$$(i) \quad (\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c}),$$

$$(ii) \quad \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{0}.$$

Relate the case $\mathbf{c} = \mathbf{a}$ and $\mathbf{d} = \mathbf{b}$ in (i) to a well-known trigonometric identity. Evaluate $(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d})$ in two ways and compare to find an explicit linear combination of $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ that is zero.

(b) Given that $[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$, show that

$$[\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}] = [\mathbf{a}, \mathbf{b}, \mathbf{c}]^2.$$

Solution. (a)(i) Using $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = \mathbf{c} \cdot (\mathbf{d} \times (\mathbf{a} \times \mathbf{b}))$ and $\mathbf{d} \times (\mathbf{a} \times \mathbf{b}) = (\mathbf{d} \cdot \mathbf{b})\mathbf{a} - (\mathbf{d} \cdot \mathbf{a})\mathbf{b}$, we obtain

$$(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c}).$$

Putting $\mathbf{c} = \mathbf{a}, \mathbf{d} = \mathbf{b}$ gives

$$|\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2|\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2,$$

equivalent to $\sin^2 \theta + \cos^2 \theta = 1$.

(a)(ii) Expand each term:

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c},$$

$$\mathbf{b} \times (\mathbf{c} \times \mathbf{a}) = (\mathbf{b} \cdot \mathbf{a})\mathbf{c} - (\mathbf{b} \cdot \mathbf{c})\mathbf{a},$$

$$\mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = (\mathbf{c} \cdot \mathbf{b})\mathbf{a} - (\mathbf{c} \cdot \mathbf{a})\mathbf{b},$$

and they cancel to give $\mathbf{0}$.

Linear combination. Compute

$$(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) = \mathbf{c} [(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{d}] - \mathbf{d} [(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}] = \mathbf{c} [\mathbf{a}, \mathbf{b}, \mathbf{d}] - \mathbf{d} [\mathbf{a}, \mathbf{b}, \mathbf{c}].$$

Also

$$(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) = -(\mathbf{c} \times \mathbf{d}) \times (\mathbf{a} \times \mathbf{b}) = \mathbf{b}[\mathbf{c}, \mathbf{d}, \mathbf{a}] - \mathbf{a}[\mathbf{c}, \mathbf{d}, \mathbf{b}].$$

Using cyclic symmetry of $[\cdot, \cdot, \cdot]$ and equating gives

$$[\mathbf{b}, \mathbf{c}, \mathbf{d}]\mathbf{a} - [\mathbf{a}, \mathbf{c}, \mathbf{d}]\mathbf{b} + [\mathbf{a}, \mathbf{b}, \mathbf{d}]\mathbf{c} - [\mathbf{a}, \mathbf{b}, \mathbf{c}]\mathbf{d} = \mathbf{0}.$$

(b)

$$[\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}] = (\mathbf{a} \times \mathbf{b}) \cdot ((\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a})).$$

But

$$(\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a}) = \mathbf{c}[(\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a}] - \mathbf{a}[(\mathbf{b} \times \mathbf{c}) \cdot \mathbf{c}] = \mathbf{c}[\mathbf{a}, \mathbf{b}, \mathbf{c}],$$

so

$$[\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}] = [\mathbf{a}, \mathbf{b}, \mathbf{c}] (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = [\mathbf{a}, \mathbf{b}, \mathbf{c}]^2.$$

Problem 12 — Solving linear vector equations with $\times$ and $\cdot$

Exact statement. Let $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ be fixed vectors in three dimensions. For each of the following equations, find all solutions for $\mathbf{r}$:

$$(i) \mathbf{r} + \mathbf{r} \times \mathbf{d} = \mathbf{c}; \quad (ii) \mathbf{r} + (\mathbf{r} \cdot \mathbf{a})\mathbf{b} = \mathbf{c}.$$

[In (ii), consider separately the cases $\mathbf{a} \cdot \mathbf{b} \neq -1$ and $\mathbf{a} \cdot \mathbf{b} = -1$].

Solution. (i) A closed-form solution is

$$\mathbf{r} = \frac{\mathbf{c} - \mathbf{c} \times \mathbf{d} + (\mathbf{c} \cdot \mathbf{d})\mathbf{d}}{1 + |\mathbf{d}|^2}.$$

Indeed, using $(\mathbf{c} \times \mathbf{d}) \times \mathbf{d} = \mathbf{d}(\mathbf{c} \cdot \mathbf{d}) - \mathbf{c}|\mathbf{d}|^2$, one checks that $\mathbf{r} + \mathbf{r} \times \mathbf{d} = \mathbf{c}$.

(ii) Let $s = \mathbf{r} \cdot \mathbf{a}$. Then $\mathbf{r} = \mathbf{c} - s\mathbf{b}$, and dotting with $\mathbf{a}$ gives

$$s = \mathbf{c} \cdot \mathbf{a} - s(\mathbf{b} \cdot \mathbf{a}) \implies s(1 + \mathbf{a} \cdot \mathbf{b}) = \mathbf{c} \cdot \mathbf{a}.$$

If $\mathbf{a} \cdot \mathbf{b} \neq -1$, then

$$s = \frac{\mathbf{c} \cdot \mathbf{a}}{1 + \mathbf{a} \cdot \mathbf{b}}, \quad \mathbf{r} = \mathbf{c} - \frac{\mathbf{c} \cdot \mathbf{a}}{1 + \mathbf{a} \cdot \mathbf{b}} \mathbf{b}.$$

If $\mathbf{a} \cdot \mathbf{b} = -1$, then solutions exist iff $\mathbf{c} \cdot \mathbf{a} = 0$. In that case s is arbitrary and

$$\mathbf{r} = \mathbf{c} - t\mathbf{b}, \quad t \in \mathbb{R}.$$

If $\mathbf{c} \cdot \mathbf{a} \neq 0$, there are no solutions.

Problem 13 — Spherical unit vectors form an orthonormal right-handed frame

Exact statement. The vectors $\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\phi$ are defined in terms of the standard basis vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$ by

$$\mathbf{e}_r = \cos \phi \sin \theta \mathbf{i} + \sin \phi \sin \theta \mathbf{j} + \cos \theta \mathbf{k},$$

$$\mathbf{e}_\theta = \cos \phi \cos \theta \mathbf{i} + \sin \phi \cos \theta \mathbf{j} - \sin \theta \mathbf{k},$$

$$\mathbf{e}_\phi = -\sin \phi \mathbf{i} + \cos \phi \mathbf{j},$$

where θ and ϕ are real numbers. Show, as efficiently as possible, that $\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\phi$ are an orthonormal right-handed set.

Solution. In components,

$$\mathbf{e}_r = (\cos \phi \sin \theta, \sin \phi \sin \theta, \cos \theta), \quad \mathbf{e}_\theta = (\cos \phi \cos \theta, \sin \phi \cos \theta, -\sin \theta), \quad \mathbf{e}_\phi = (-\sin \phi, \cos \phi, 0).$$

Unit lengths.

$$|\mathbf{e}_r|^2 = \sin^2 \theta (\cos^2 \phi + \sin^2 \phi) + \cos^2 \theta = 1, \quad |\mathbf{e}_\theta|^2 = \cos^2 \theta (\cos^2 \phi + \sin^2 \phi) + \sin^2 \theta = 1,$$

$$|\mathbf{e}_\phi|^2 = \sin^2 \phi + \cos^2 \phi = 1.$$

Orthogonality.

$$\mathbf{e}_r \cdot \mathbf{e}_\theta = \sin \theta \cos \theta (\cos^2 \phi + \sin^2 \phi) - \sin \theta \cos \theta = 0,$$

$$\mathbf{e}_r \cdot \mathbf{e}_\phi = \cos \phi \sin \theta (-\sin \phi) + \sin \phi \sin \theta (\cos \phi) = 0,$$

$$\mathbf{e}_\theta \cdot \mathbf{e}_\phi = \cos \phi \cos \theta (-\sin \phi) + \sin \phi \cos \theta (\cos \phi) = 0.$$

Hence the set is orthonormal.

Right-handedness.

$$\mathbf{e}_\theta \times \mathbf{e}_\phi = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos \phi \cos \theta & \sin \phi \cos \theta & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{vmatrix} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) = \mathbf{e}_r.$$

Therefore $\mathbf{e}_\theta \times \mathbf{e}_\phi = \mathbf{e}_r$, so $(\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\phi)$ is a right-handed orthonormal set.

Vector diagrams for Problems 1–13

Problem 1: if $|z - (1 + i)| < 1$ then $\sqrt{5} - 1 < |z - 3| < \sqrt{5} + 1$.

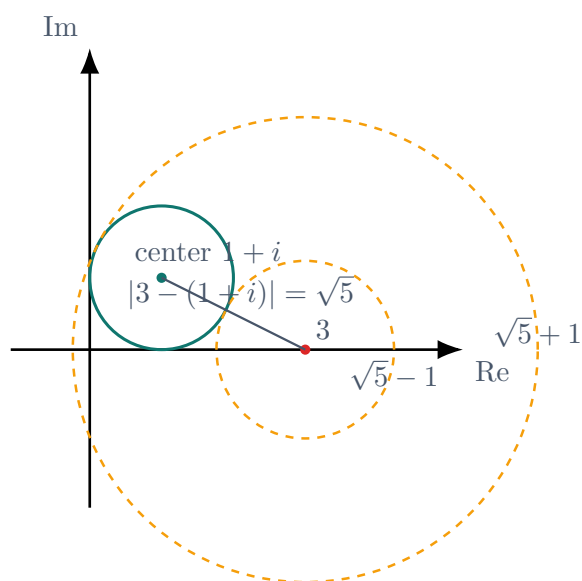


Figure 6: Problem 1: disk and distance bounds shown as an annulus around 3.

Problem 2(a): modulus-argument pictures for $1 + z$ and $1 - \bar{z}$.

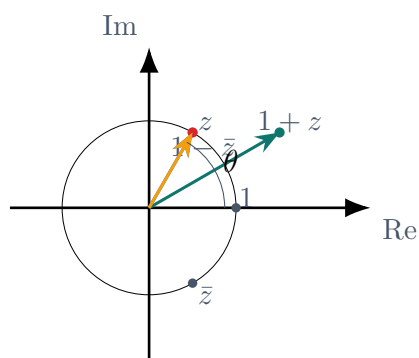


Figure 7: Problem 2(a): unit-circle geometry for z , $1 + z$, and $1 - \bar{z}$.

Problem 2(b): $w^2 \in i\mathbb{R} \Rightarrow w$ lies on two lines.

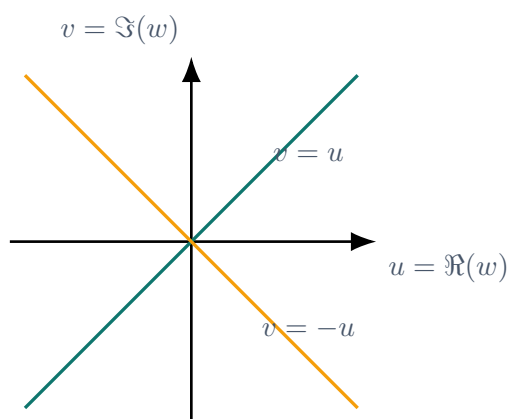


Figure 8: Problem 2(b): locus of w is the pair of lines $v = \pm u$.

Problem 3: medians concur at the centroid $G = (z_1 + z_2)/3$.

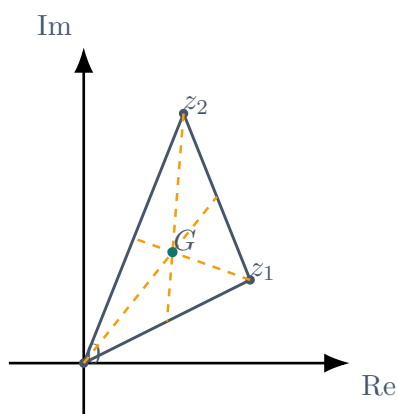


Figure 9: Problem 3: medians and centroid for the triangle with vertices $0, z_1, z_2$.

Problem 4: conjugate pairs (ω, ω^4) and (ω^2, ω^3) .

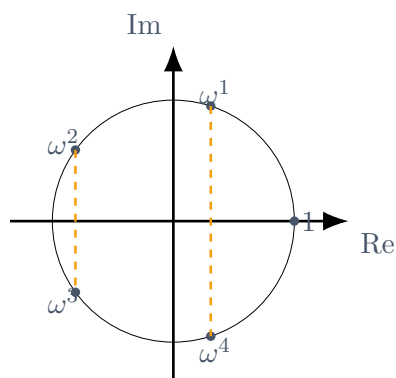


Figure 10: Problem 4: the 5th roots of unity on the unit circle, with conjugate pairings.

Problem 5: $z = \frac{\pi}{2} + 2\pi n \pm i \ln(2 + \sqrt{3})$.

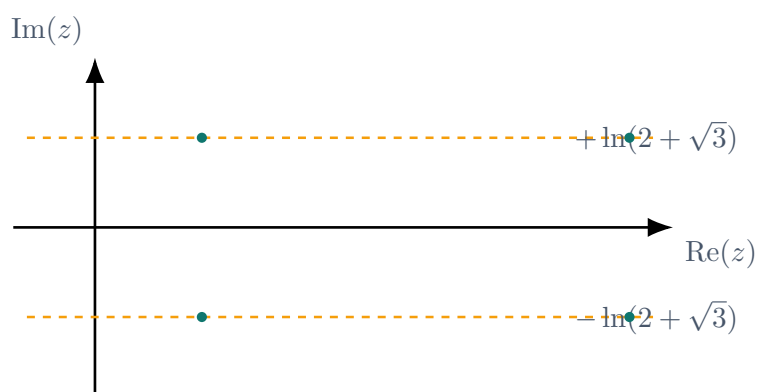


Figure 11: Problem 5: solutions of $\sin z = 2$ lie on two horizontal lines in the complex plane.

Problem 6: example $p = 1$, $\lambda = 2$, $0.6 = 0.6$ (circles meet orthogonally).

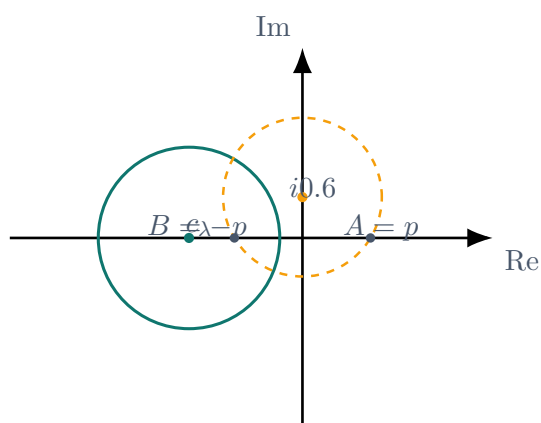


Figure 12: Problem 6: example of C_λ and S_μ (orthogonal intersection).

Problem 7: altitudes concur at the orthocenter H .

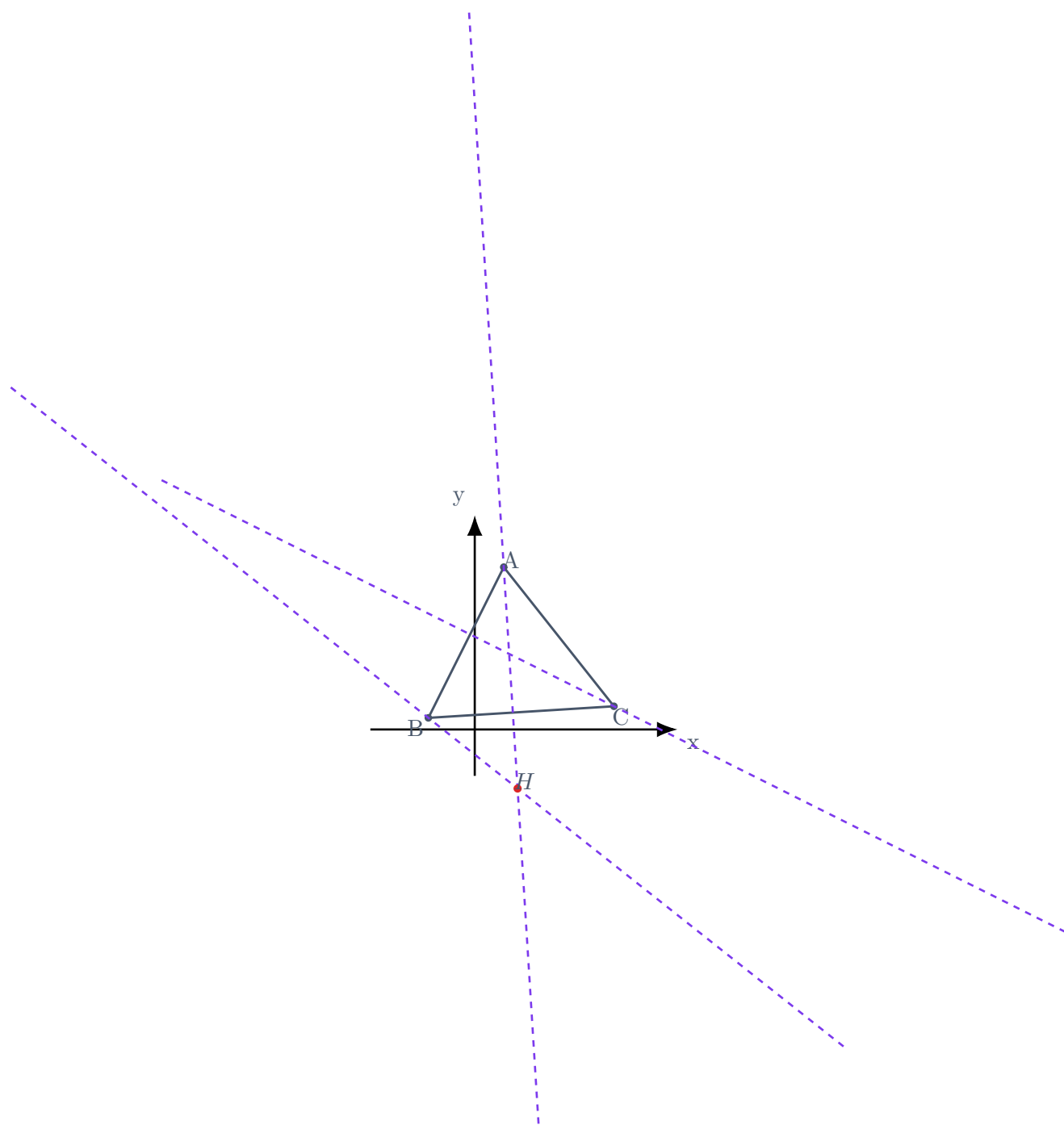


Figure 13: Problem 7: triangle altitudes meeting at the orthocenter.

Problem 8: $a = x + (x \cdot y) y$ (schematic).

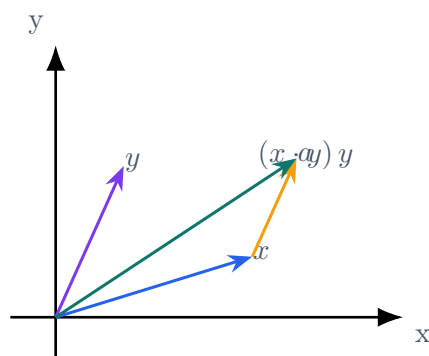


Figure 14: Problem 8: schematic vector addition for $a = x + (x \cdot y) y$.

Problem 9: $u \times v = v \times w = w \times u$ (same directed area).

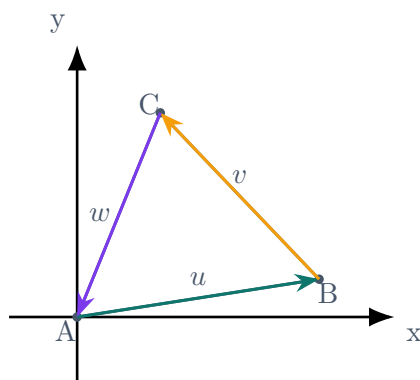


Figure 15: Problem 9: triangle side vectors and the equal cross-product (area-vector) idea.

Problem 10: affine combinations describe the line AB and plane ABC .

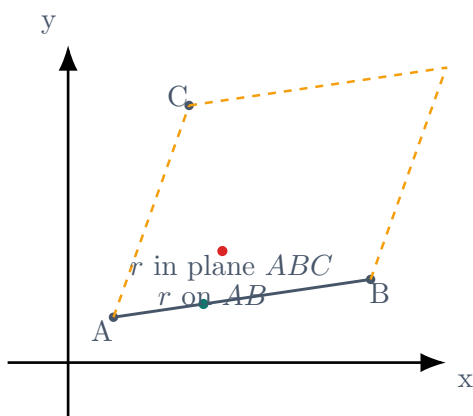


Figure 16: Problem 10: geometric meaning of affine combinations (line through A, B and plane through A, B, C).

Problem 11: cross-product area and triple-product volume (schematic).

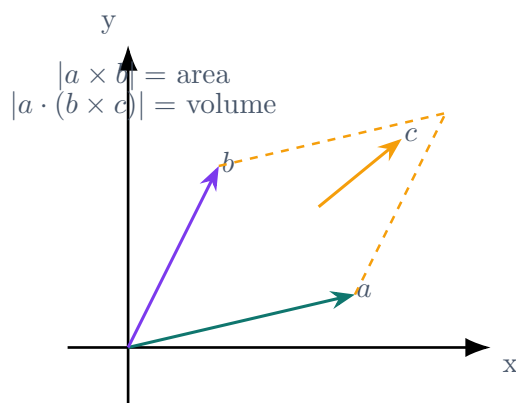


Figure 17: Problem 11: schematic for cross-product area and scalar triple-product volume.

Problem 12: two linear vector equations (schematic).

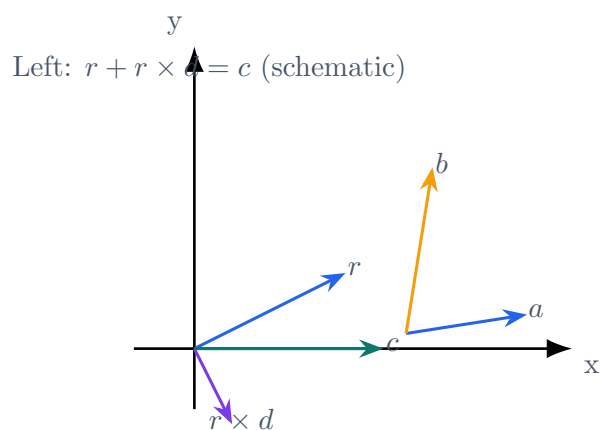


Figure 18: Problem 12: schematics for the two vector equations.

Problem 13: orthonormal, right-handed frame ($e_3 \times e_1 = e_2$).

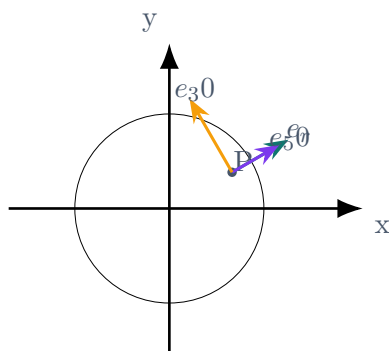


Figure 19: Problem 13: spherical basis vectors (2D projection for illustration).